# Infectious Diseases OSCE Cases

Question and answer review by exam year

Key: Drug names **Critical point** ★ Repeated case Answers follow IDSA, ATS, CDC and SHEA guidance.

## 2025

### Case 1: Hospital-Acquired Pneumonia (HAP) on Hemodialysis ★ Repeated case, also in 2018–2019

**Case details:** Hospitalized 4 days. Hemodialysis 3 times/week. Cough, SOB, fever; diagnosed with pneumonia. Unit antibiogram: MRSA >42%; Pseudomonas pan-sensitive except FQ resistance >30%. Home: Valproic acid (epilepsy), Citalopram (depression).

| Question | Answer |
|---|---|
| **Q1. What is the diagnosis and why?** | • HAP: pneumonia that starts ≥48 hours after admission and was not incubating at admission<br>• Not VAP: patient is not intubated<br>• Not CAP/HCAP: HCAP was retired in 2016; hemodialysis alone is not an MDR risk factor for HAP<br>• Supporting data: fever, cough, SOB, new infiltrate on chest X-ray, leukocytosis, hypoxemia |
| **Q2. What will you cover?** | • MRSA: unit MRSA prevalence is >20% (high-risk threshold; the 42% here is well above it)<br>• Pseudomonas aeruginosa and other gram-negative bacilli, plus MSSA<br>• Two antipseudomonal agents are not required here: not ventilated, no septic shock, no structural lung disease<br>• Do not add a fluoroquinolone as the second agent: local resistance >30% (avoid if >10%) |
| **Q3. What specific agent will you use?** | • Anti-MRSA: Vancomycin IV (Linezolid is not suitable, see Q7)<br>• Antipseudomonal β-lactam: Piperacillin-tazobactam 2.25 g IV q8h plus 0.75 g after each dialysis (nosocomial pneumonia dosing in HD)<br>• Alternative β-lactam: Cefepime 1 g IV q24h after HD (renal adjustment of 2 g q8h; neurotoxicity risk in ESRD)<br>• Avoid Meropenem/Imipenem: lowers Valproic acid levels (see Q8)<br>• Monitor: temperature, WBC, oxygenation, respiratory rate at 48–72 h; culture and MRSA nasal PCR<br>• Non-pharm: head of bed 30–45°, oral care, aspiration precautions, early mobilization, hand hygiene |
| **Q4. LD and maintenance dose of Vancomycin on HD, and timing relative to dialysis?** | • Loading dose: 25 mg/kg actual body weight (range 20–35, max ~3 g), given after HD or during the last hour of dialysis<br>• Maintenance: 5–10 mg/kg after each HD session (about 500–1000 mg)<br>• Why after HD: high-flux membranes remove Vancomycin, so give it when dialysis ends<br>• Dose by level: re-dose according to the pre-HD concentration |
| **Q5. When to take the level (on HD)?** | • Pre-dialysis level: draw just before the next HD session<br>• Avoid levels right after HD (rebound redistribution; wait ≥4–6 hours if a post-HD level is needed)<br>• Repeat if dialysis prescription or residual function changes |
| **Q6. Target level?** | • Preferred: AUC/MIC 400–600 mg·h/L (Bayesian or two-level AUC); trough-only monitoring is no longer recommended for serious MRSA<br>• Practical HD surrogate: pre-dialysis concentration 15–20 mg/L (older target used in HD units) |
| **Q7. Why not Linezolid or Daptomycin?** | • Linezolid + Citalopram: weak MAO inhibition → **serotonin syndrome**; also thrombocytopenia is worse in renal failure<br>• Daptomycin: inactivated by pulmonary surfactant → not for pneumonia<br>• Vancomycin is the best choice for MRSA HAP here |
| **Q8. Why not a fluoroquinolone or Meropenem?** | • Fluoroquinolone: local Pseudomonas resistance >30%; also QT risk with Citalopram<br>• Meropenem: carbapenems drop Valproic acid concentrations by 60–100% within 24 hours → breakthrough seizures; the effect is not overcome by raising the Valproic acid dose<br>• Choose: Piperacillin-tazobactam or Cefepime (no Valproic acid interaction) |
| **Q9. Patient developed CRAP: what management option?** | • Treatment failure or respiratory failure (no improvement at 48–72 h, worsening oxygenation):<br>• Reassess: diagnosis, complications (empyema, abscess, ARDS), alternative sources, adherence to dosing<br>• Escalate: review cultures/sensitivities; broaden to Meropenem or add a second antipseudomonal agent only after the Valproic acid issue is managed<br>• Respiratory support as needed; drain any effusion |
| **Q10. Minimum duration for treatment?** | • 7 days (IDSA/ATS 2016 and ACCP) if the patient responds<br>• Stop MRSA therapy early if nasal PCR and cultures are negative |
| **Also be ready for (exam add-ons)** | • Goals: eradicate the pathogen, reduce mortality, stop therapy at the shortest effective duration (7 days), avoid toxicity and resistance<br>• Non-pharm: head of bed 30–45°, early mobilization, oral care, aspiration precautions, pulmonary toilet, isolate MDR organisms, remove unnecessary devices<br>• First-line: empiric per risk: MRSA coverage (Vancomycin 25–30 mg/kg load by AUC, or Linezolid 600 mg q12h) + antipseudomonal β-lactam (Cefepime 2 g q8h, Piperacillin-tazobactam 4.5 g q6h, or Meropenem |

| Question | Answer |
|---|---|
| | 1 g q8h); hemodialysis: Vancomycin 25 mg/kg load then 500–1000 mg after each session by pre-dialysis level<br>• Pregnancy: β-lactams and Vancomycin safe; avoid fluoroquinolones, tetracyclines, aminoglycosides; Linezolid only if essential<br>• Renal/dialysis: Cefepime reduce (neurotoxicity); Meropenem 500 mg q24h if CrCl <10; Levofloxacin 750 mg q48h; Vancomycin dosed after HD<br>• Liver: Linezolid/Tigecycline caution; Metronidazole reduce; Azithromycin caution<br>• Interactions: Linezolid + SSRIs (Citalopram) = serotonin syndrome; Valproic acid + Carbapenems = loss of seizure control (avoid Meropenem); fluoroquinolone resistance high<br>• Counseling: Linezolid: avoid tyramine-rich foods, thrombocytopenia (CBC weekly), serotonin syndrome; Vancomycin: infusion reaction (≤1 g/h), kidney/hearing monitoring<br>• Monitoring: temperature, WBC, oxygenation, cultures, Vancomycin AUC/trough pre-HD, SCr, platelets |

## Case 2: Tuberculosis with Aspergillosis ★ Repeated case, also in 2022–2023

**Case details:** Pulmonary TB with aspergillosis. Treated with first-line anti-TB therapy; later hepatitis and a request to start Voriconazole.

| Question | Answer |
|---|---|
| **Q1. What are the agents used for TB treatment?** | • Intensive phase (2 months): Isoniazid, Rifampin, Pyrazinamide, Ethambutol<br>• Continuation phase (4 months): Isoniazid + Rifampin<br>• Add: Pyridoxine with Isoniazid |
| **Q2. What are the doses for each agent?** | • Isoniazid: 5 mg/kg daily (max 300 mg)<br>• Rifampin: 10 mg/kg daily (max 600 mg)<br>• Pyrazinamide: 20–25 mg/kg daily (about 1500 mg for 56–75 kg)<br>• Ethambutol: 15–20 mg/kg daily (about 1200 mg for 56–75 kg)<br>• Pyridoxine: 25–50 mg daily<br>• Administration: daily, on an empty stomach, directly observed therapy |
| **Q3. Would you add a supplement to the regimen?** | • Yes: Pyridoxine (vitamin B6) 25–50 mg daily<br>• Why: Isoniazid depletes Pyridoxine → peripheral neuropathy and neurotoxicity<br>• Highest risk: diabetes, alcohol use, HIV, malnutrition, renal failure, pregnancy |
| **Q4. What is the duration of anti-TB treatment?** | • Standard pulmonary TB: 6 months (2 months RIPE + 4 months HR)<br>• Extend to 9 months if cavitation and sputum culture still positive at 2 months<br>• Based on doses taken, not calendar time<br>• CNS TB: 9–12 months; bone/joint: 6–9 months<br>• Shorter option (CDC 2022): 4-month Rifapentine + Isoniazid + Pyrazinamide + Moxifloxacin for drug-susceptible TB in patients ≥12 years; MDR/RR-TB: 6-month BPaLM (Bedaquiline, Pretomanid, Linezolid, Moxifloxacin) |
| **Q5. Developed hepatitis (liver enzymes 4× normal). What will you do? Stop what, and when to re-challenge?** | • Check symptoms and exclude other causes: alcohol, Acetaminophen, viral hepatitis, herbals<br>• Stop thresholds: ALT ≥5× ULN with or without symptoms, or ≥3× ULN with symptoms (nausea, vomiting, jaundice, abdominal pain)<br>• At 4× ULN: **symptomatic → stop all hepatotoxic drugs**; asymptomatic → may continue with weekly LFTs<br>• If stopped: hold Isoniazid, Rifampin, Pyrazinamide; keep a non-hepatotoxic bridge, eg Ethambutol + Levofloxacin or Moxifloxacin<br>• Re-challenge when ALT <2× ULN (and bilirubin normal): restart sequentially<br>  ◦ Rifampin first, then Isoniazid after about 1 week, then Pyrazinamide after another week<br>  ◦ Pyrazinamide is the one most often left out permanently (extend therapy to 9 months)<br>• Monitor: LFTs and bilirubin weekly until normal, then every 2 weeks; symptoms at each visit |
| **Q6. The team decided to start Voriconazole. What is your response?** | • Do not combine: Rifampin is a strong CYP3A4/2C19 inducer → Voriconazole exposure falls about 95% → treatment failure (contraindicated)<br>• Isavuconazole and Posaconazole are also unsuitable with Rifampin<br>• Switch to: liposomal Amphotericin B 3–5 mg/kg IV daily<br>• Other option: replace Rifampin with Rifabutin (needs Voriconazole level monitoring), only if TB regimen allows<br>• Duration: 6–12 weeks, guided by clinical and radiologic response |
| **Q7. The team decided to start Amphotericin. What will you monitor? (Include infusion reactions.)** | • Infusion reactions: fever, chills, rigors, hypotension, dyspnea<br>  ◦ Premedicate with Acetaminophen and Diphenhydramine ± Hydrocortisone; Meperidine for rigors<br>  ◦ Infuse over 2–4 hours and slow down if reactions occur<br>• Nephrotoxicity: SCr, BUN, urine output daily; give 500–1000 mL normal saline before each dose<br>• Electrolytes: K and Mg wasting (daily, replace), Na, bicarbonate<br>• CBC: normochromic normocytic anemia<br>• LFTs, phlebitis at the infusion site<br>• Response: fever, imaging, galactomannan |
| **Also be ready for (exam add-ons)** | • Goals: cure, prevent transmission, prevent resistance and relapse, minimize hepatotoxicity<br>• Non-pharm: airborne isolation until 3 negative smears (or 2 weeks effective therapy and clinical improvement), contact tracing, DOT (directly observed therapy), nutrition, stop alcohol, HIV test |

| Question | Answer |
|---|---|
| | • First-line: RIPE: Rifampin 10 mg/kg (600 mg), Isoniazid 5 mg/kg (300 mg), Pyrazinamide 25 mg/kg, Ethambutol 15–20 mg/kg × 2 months then Rifampin + Isoniazid × 4 months (6 months total); add Pyridoxine 25–50 mg daily<br>• Pregnancy: RIPE with Pyridoxine is used (Pyrazinamide is recommended by WHO but omitted by US guidance unless HIV or resistance; without it, give 9 months of Isoniazid + Rifampin + Ethambutol); avoid Streptomycin (fetal ototoxicity); breastfeeding is safe<br>• Renal/dialysis: Isoniazid and Rifampin no adjustment; Pyrazinamide and Ethambutol 3 times weekly (25 mg/kg and 15–25 mg/kg after dialysis) if CrCl <30<br>• Liver: hold if ALT ≥5× ULN (≥3× with symptoms) or bilirubin rise; re-challenge Rifampin first then Isoniazid then Pyrazinamide after ALT <2× ULN; alternatives: Levofloxacin, Ethambutol, Amikacin<br>• Interactions: Rifampin induces CYP3A4: Warfarin, DOACs, Tacrolimus, antiretrovirals (use Rifabutin or adjust Dolutegravir to 50 mg BID), azoles (Voriconazole: use Amphotericin), oral contraceptives, steroids<br>• HIV: start ART 2 weeks after TB therapy if CD4 <50 (and within 8 weeks if ≥50); Dolutegravir 50 mg BID with Rifampin<br>• Counseling: take on an empty stomach, adherence for the full course, orange body fluids (Rifampin), vision changes (Ethambutol: stop and call), jaundice/dark urine/nausea (stop and call), gout (Pyrazinamide), neuropathy (Isoniazid: Pyridoxine), avoid alcohol; hormonal contraceptives fail<br>• Monitoring: LFTs baseline, 2, 4, 8 weeks (monthly if risk), visual acuity and color vision monthly (Ethambutol), uric acid, sputum at 2 months, weight |

## Case 3: Peritonitis and C. difficile Infection

**Case details:** Medically free patient admitted for appendectomy; 3 days after surgery developed peritonitis. T 39 °C, WBC 16.

| Question | Answer |
|---|---|
| **Q1. What is the classification of peritonitis (and based on what)?** | • Secondary peritonitis, hospital-acquired (health care-associated)<br>• Basis: onset after 48 hours in hospital, postoperative (3 days after appendectomy), likely leak or abscess<br>• Implication: higher risk of resistant organisms than community-acquired |
| **Q2. What microbes will you cover?** | • Enteric gram-negative bacilli: E. coli, Klebsiella, Enterobacter, Pseudomonas (hospital-acquired)<br>• Anaerobes: Bacteroides fragilis<br>• Gram-positive: Streptococcus, Enterococcus (consider with postoperative infection)<br>• Add if risk factors: MRSA (colonized or prior failure), Candida |
| **Q3. Give one combination regimen and one monotherapy.** | • Combination: Cefepime 2 g IV q8h + Metronidazole 500 mg IV q8h<br>• Monotherapy: Piperacillin-tazobactam 4.5 g IV q6h, or Meropenem 1 g IV q8h<br>• Avoid Ciprofloxacin or Levofloxacin + Metronidazole for health care-associated infection<br>• Source control: drain/operate within 6–12 hours<br>• Duration: about 4 days after adequate source control (STOP-IT); longer only if source control is inadequate<br>• Monitor: temperature, WBC, abdominal exam, drain output, SCr, culture results |
| **Q4. After 3 days the patient developed C. difficile with megacolon and high WBC. What is the type of CDI?** | • Fulminant CDI: hypotension/shock, ileus or megacolon<br>• Severity scale: non-severe WBC <15,000 and SCr ≤1.5; severe WBC ≥15,000 or SCr >1.5 |
| **Q5. What is the management of this type? Doses, route, duration.** | • Vancomycin 500 mg PO/NG q6h × 10 days<br>• Plus Metronidazole 500 mg IV q8h (especially with ileus)<br>• Stop the inciting antibiotic if possible and stop PPIs if not needed<br>• Surgical consult: colectomy for perforation, toxic megacolon or worsening shock<br>• Non-pharm: contact precautions, soap and water hand washing, dedicated equipment, fluid resuscitation<br>• Monitor: stool frequency, abdominal exam/imaging, WBC, lactate, SCr, K, hydration<br>• Non-fulminant initial CDI: Fidaxomicin 200 mg BID × 10 days is preferred (IDSA/SHEA 2021); Vancomycin 125 mg PO QID is an acceptable alternative |
| **Q6. What modification will you do if the patient developed ileus?** | • Add rectal Vancomycin: 500 mg in 100 mL normal saline as a retention enema q6h<br>• Continue: oral/NG Vancomycin 500 mg q6h and IV Metronidazole 500 mg q8h<br>• Stop the enema and Metronidazole when the ileus resolves |
| **Q7. Peritonitis culture showed Candida krusei. What will you give?** | • Echinocandin: Micafungin 100 mg IV daily (or Caspofungin 70 mg load then 50 mg daily, or Anidulafungin 200 mg load then 100 mg daily)<br>• Do not use Fluconazole: C. krusei is intrinsically resistant<br>• Alternative: liposomal Amphotericin B 3–5 mg/kg daily (more toxic)<br>• Duration: at least 2 weeks after source control and clinical resolution; 14 days after first negative culture if candidemia<br>• Monitor: blood cultures, fever, LFTs (echinocandins), SCr |
| **Also be ready for (exam add-ons)** | • Goals: control the source, eradicate the pathogens, treat CDI to prevent recurrence and colectomy<br>• Non-pharm: source control (drainage/surgery), fluids, stop the offending antibiotic and PPI if possible, contact precautions and soap-and-water hand hygiene, avoid antimotility drugs<br>• First-line: hospital-acquired peritonitis: Piperacillin-tazobactam 4.5 g q6h (monotherapy) or Cefepime 2 g q8h + Metronidazole 500 mg q8h; fulminant CDI: Vancomycin 500 mg PO/NG q6h + Metronidazole 500 |

| Question | Answer |
|---|---|
| | mg IV q8h (+ Vancomycin PR 500 mg in 100 mL q6h if ileus); Candida krusei: Micafungin 100 mg or Caspofungin (Fluconazole-resistant)<br>• Pregnancy: Piperacillin-tazobactam and Meropenem acceptable; oral Vancomycin or Fidaxomicin for CDI; Metronidazole avoided early first trimester; avoid Fluconazole (first trimester), use Amphotericin B if needed<br>• Renal/dialysis: adjust β-lactams (Pip-tazo 2.25 g q8h if CrCl <20); oral Vancomycin no adjustment; Micafungin none<br>• Liver: Metronidazole reduce dose in Child-Pugh C; Caspofungin reduce maintenance dose to 35 mg in Child-Pugh B<br>• GI: stop PPI; CDI recurrence: Fidaxomicin 200 mg BID × 10 days, extended-pulsed, or Bezlotoxumab (anti-toxin B monoclonal antibody, no longer marketed since 2025); microbiome products (Rebyota rectal, Vowst oral) after antibiotic completion<br>• Counseling: hand washing with soap and water, finish the course, hydration, report worsening diarrhea/distension, avoid unnecessary antibiotics/PPI, no probiotics routinely |

# 2024

## Case 1: Pyelonephritis in Pregnancy, ESBL, then Carbapenem-Resistant Acinetobacter (CRAB) Pneumonia

**Case details:** Pregnant woman in the outpatient clinic with nausea, vomiting and flank pain. Urinalysis: nitrite positive and leukocytes; fever and high WBC. She is admitted without improvement; culture grows ESBL. On day 2 of Meropenem she develops cough and pneumonia: culture shows carbapenem-resistant Acinetobacter.

| Question | Answer |
|---|---|
| **Q1. What is the diagnosis?** | • **Acute complicated pyelonephritis in pregnancy** (admit for IV therapy) |
| **Q2. What are the signs and symptoms?** | • Fever, flank pain and costovertebral tenderness, nausea and vomiting, dysuria and urgency, pyuria, nitrite, bacteriuria; look for sepsis and preterm contractions |
| **Q3. What is the treatment?** | • **IV β-lactam:** Ceftriaxone 1–2 g q24h until afebrile 24–48 h, then oral step-down by culture (Cephalexin, Cefpodoxime, Amoxicillin-clavulanate) to complete **10–14 days**<br>• Avoid fluoroquinolones, tetracyclines and TMP-SMX (first trimester); give a suppressive prophylaxis afterward (Cephalexin 250–500 mg nightly or Nitrofurantoin until delivery) |
| **Q4. What is the most common organism?** | • **Escherichia coli** (then Klebsiella, Proteus, Enterococcus, Group B Streptococcus) |
| **Q5. Culture shows ESBL: what will you give?** | • **Carbapenem:** Meropenem 1 g IV q8h (or Ertapenem 1 g daily); acceptable in pregnancy; 7–14 days with source review |
| **Q6. Pneumonia on Day 2 with carbapenem-resistant Acinetobacter: what will you give?** | • IDSA 2024 AMR guidance: **Sulbactam-durlobactam** (+ a carbapenem) is preferred; alternatives: high-dose **Ampicillin-sulbactam** (sulbactam 9 g/day, e.g., 27 g/day total in divided doses) combined with a second agent (Polymyxin B/Colistin, Minocycline, Tigecycline, Cefiderocol)<br>• In pregnancy avoid tetracyclines (Minocycline, Tigecycline) and aminoglycosides; Colistin or Polymyxin B, Ampicillin-sulbactam and Cefiderocol are the usual options (limited data: use with infectious diseases and obstetric input) |
| **Q7. What are the side effects of Colistin?** | • **Nephrotoxicity** (AKI, dose-related) and **neurotoxicity** (paresthesia, ataxia, dizziness, neuromuscular blockade and respiratory arrest); also bronchospasm if inhaled<br>• Load 9 million IU then 4.5 million IU q12h (adjust to CrCl); Polymyxin B has less nephrotoxicity and needs no renal adjustment; monitor SCr daily |
| **Also be ready for (exam add-ons)** | • Goals: eradicate infection and sepsis risk, prevent preterm labor and fetal harm, prevent recurrence<br>• Non-pharm: hospital admission, IV fluids, antipyretics (Acetaminophen), left lateral position, fetal monitoring, urine culture each trimester and after treatment, infection control for CRAB (contact isolation)<br>• First-line: pyelonephritis: Ceftriaxone 1–2 g IV q24h then oral step-down to 10–14 days; ESBL: Meropenem 1 g q8h; CRAB: sulbactam-durlobactam (preferred) or high-dose Ampicillin-sulbactam + Polymyxin B/Colistin/Cefiderocol<br>• Pregnancy: β-lactams, carbapenems and Cephalexin are safe; avoid fluoroquinolones, tetracyclines (Minocycline, Tigecycline), TMP-SMX (first trimester and near term) and aminoglycosides (ototoxicity); give suppressive Cephalexin 250–500 mg nightly after pyelonephritis<br>• Renal/dialysis: adjust Meropenem to CrCl (CrCl 26–50: 1 g q12h); Colistin dose-adjust and watch SCr; Polymyxin B no renal adjustment but still nephrotoxic<br>• Liver: no major adjustment for β-lactams; Tigecycline is avoided anyway<br>• Counseling: complete the course, hydration, report contractions, bleeding or leaking fluid, recurrent fever; Colistin neurotoxicity (tingling, dizziness, weakness) and kidney monitoring |

## Case 2: Community-Acquired Pneumonia with Prior MRSA and QT Prolongation

**Case details:** 55-year-old man, otherwise healthy, with runny nose, sore throat, cough and fever; chest X-ray shows an infiltrate; WBC high. MRSA pneumonia 2

months ago, treated successfully. Takes a calcium supplement. **QTc 527 ms.**

| Question | Answer |
|---|---|
| **Q1. What type of pneumonia is this?** | • **Community-acquired pneumonia** with a risk factor for MRSA (prior MRSA isolation): IDSA/ATS 2019 adds MRSA coverage and sends cultures and nasal PCR |
| **Q2. Which medicines should you avoid? Mention 2 classes.** | • **Macrolides** (Azithromycin, Clarithromycin) and **fluoroquinolones** (Levofloxacin, Moxifloxacin): QT prolongation; also avoid other QT drugs (Ondansetron, antipsychotics, azoles) |
| **Q3. What will you give empirically? Why? Which organisms?** | • **Ceftriaxone** 1–2 g IV q24h (S. pneumoniae, H. influenzae) + **Doxycycline** 100 mg PO/IV q12h (atypicals: Mycoplasma, Chlamydophila, Legionella: the sore throat points to atypical/viral cause) + **Vancomycin** (MRSA)<br>• Vancomycin 25 mg/kg load then 15 mg/kg q12h, adjusted by AUC 400–600<br>• Newer options when macrolides and fluoroquinolones are unsuitable: Omadacycline 200 mg IV on day 1 then 100 mg daily (covers atypicals and MRSA, no QT effect); Lefamulin 150 mg IV q12h (avoid here: QT prolongation) |
| **Q4. Side effects of the chosen medicines?** | • Ceftriaxone: rash, diarrhea/C. difficile, biliary sludge; Doxycycline: GI upset, esophagitis, photosensitivity; Vancomycin: nephrotoxicity, infusion reaction, ototoxicity |
| **Q5. Counseling points?** | • Doxycycline with a full glass of water, upright for 30 min, sun protection, **separate from the calcium supplement by 2–3 h**; complete the course; report diarrhea, rash, hearing changes or low urine output |
| **Q6. What are the monitoring parameters?** | • Temperature, WBC, oxygenation, chest X-ray; SCr and urine output, Vancomycin levels (AUC); **ECG/QTc**, K and Mg; nasal MRSA PCR to de-escalate |
| **Q7. Vancomycin is out of stock: what will you give, dose and frequency?** | • **Linezolid 600 mg PO/IV q12h** (100% oral bioavailability; no QT effect) |
| **Q8. Monitoring for Linezolid?** | • CBC weekly (thrombocytopenia, anemia), serotonin syndrome with SSRIs/SNRIs, lactic acidosis, visual changes and neuropathy with >28 days, BP and tyramine foods; duration 5–7 days |
| **Q9. QTc improved and the team wants a fluoroquinolone: any interaction with his home medicine? Mechanism?** | • **Calcium (divalent cation) chelates the fluoroquinolone** in the gut → lower absorption and treatment failure<br>• Give Levofloxacin ≥2 h before or after calcium (Ciprofloxacin: 2 h before or 6 h after; Moxifloxacin: 4 h before or 8 h after)<br>• A fluoroquinolone still prolongs QT: repeat ECG and keep K and Mg normal; avoid if QTc stays >500 ms |
| **Also be ready for (exam add-ons)** | • Goals: cure the infection, prevent complications, minimize resistance and QT/drug interactions, 5-day course<br>• Non-pharm: severity assessment (CURB-65, IDSA/ATS), oxygen, hydration, rest, smoking cessation, vaccines (pneumococcal, influenza, COVID-19, RSV), follow-up chest X-ray only if indicated<br>• First-line: inpatient non-severe: Ceftriaxone 1–2 g daily + Azithromycin 500 mg daily (or Doxycycline 100 mg BID); with penicillin allergy: respiratory fluoroquinolone (Levofloxacin 750 mg daily or Moxifloxacin 400 mg) or Doxycycline; severe: β-lactam + macrolide; add MRSA/Pseudomonas coverage only with risk factors (Vancomycin or Linezolid; Cefepime/Pip-tazo)<br>• Pregnancy: Amoxicillin, Ceftriaxone, Azithromycin safe; avoid Doxycycline and fluoroquinolones; Pneumococcal/influenza vaccines safe<br>• Renal/dialysis: Levofloxacin 750 mg q48h if CrCl 20–49; Gentamicin extended-interval by CrCl with levels; Azithromycin no change<br>• Liver: Azithromycin/Moxifloxacin caution in severe disease<br>• Interactions: Azithromycin/quinolones QT prolongation (stop and use Doxycycline); Doxycycline with ferrous sulfate/antacids (separate by 2–3 h); Levofloxacin tendon risk with steroids; ofloxacin anaphylaxis: avoid all fluoroquinolones<br>• Counseling: complete the course, Doxycycline: sun protection, take upright with water, avoid dairy/iron close to the dose; Azithromycin: palpitations; Levofloxacin: tendon pain, dysglycemia<br>• Gentamicin TDM: extended-interval 5–7 mg/kg q24h (trough <1 mg/L), conventional peak 5–10, trough <2; SCr, hearing/vestibular function<br>• QT: avoid macrolides, fluoroquinolones, Ondansetron, antipsychotics and azoles when QTc >500 ms; keep K ≥4 and Mg ≥2; repeat ECG at 24–48 h; Doxycycline and β-lactams have no QT effect |

# 2022–2023

## Case 1: Anti-TB Therapy and Toxicities ★ Repeated case, also in 2025

**Case details:** Pulmonary TB started on four first-line anti-TB medications.

| Question | Answer |
|---|---|
| **Q1. Vision problems: which medication, and how to adjust?** | • Ethambutol (optic neuritis, dose and duration related, esp. >15–25 mg/kg or renal impairment)<br>• Action: stop Ethambutol immediately and refer to ophthalmology<br>• If TB isolate is known susceptible: continue Isoniazid + Rifampin + Pyrazinamide; otherwise substitute Moxifloxacin or Levofloxacin<br>• If vision does not recover: Isoniazid optic neuritis is rare but possible, so stop Isoniazid<br>• Prevention: baseline Snellen and color vision (Ishihara), monthly visual checks, renal dose adjustment |

| Question | Answer |
|---|---|
| **Q2. Gout: which medication, and how to adjust?** | • Pyrazinamide: inhibits renal urate excretion → hyperuricemia<br>• Asymptomatic hyperuricemia: no change, monitor uric acid<br>• Acute gout: Colchicine or Prednisone (watch hepatic effect of NSAIDs with TB drugs)<br>• Do not reduce the Pyrazinamide dose: it is exposure-dependent for sterilizing activity; stop it only for severe or recurrent gout and extend therapy to 9 months<br>• Urate-lowering therapy: per usual gout criteria (eg after a second attack)<br>• Monitor: uric acid, joint symptoms, LFTs |
| **Q3. Liver enzymes >3×: which medication, and how to adjust?** | • Culprits: Isoniazid and Pyrazinamide (hepatocellular, ALT/AST); Rifampin (cholestatic, ALP/bilirubin)<br>• Stop all hepatotoxic drugs: ALT ≥5× ULN (any symptoms) or ≥3× ULN with symptoms<br>• Re-challenge when ALT <2× ULN and bilirubin <2 mg/dL: Rifampin first (1 week), then Isoniazid (1 week), then Pyrazinamide<br>• Bridge meanwhile: Ethambutol + a fluoroquinolone ± aminoglycoside<br>• Monitor: LFTs and bilirubin weekly until stable; exclude alcohol, Acetaminophen, viral hepatitis |
| **Q4. SCr rose by >3: how to adjust Rifampin?** | • No adjustment for Rifampin (and Isoniazid): hepatic clearance<br>• Adjust Pyrazinamide and Ethambutol when CrCl <30 or on HD: 25–35 mg/kg (Pyrazinamide) and 15–25 mg/kg (Ethambutol) three times weekly, after dialysis<br>• Monitor: SCr, visual acuity, uric acid |
| **Also be ready for (exam add-ons)** | • Goals: cure, prevent transmission, prevent resistance and relapse, minimize hepatotoxicity<br>• Non-pharm: airborne isolation until 3 negative smears (or 2 weeks effective therapy and clinical improvement), contact tracing, DOT (directly observed therapy), nutrition, stop alcohol, HIV test<br>• First-line: RIPE: Rifampin 10 mg/kg (600 mg), Isoniazid 5 mg/kg (300 mg), Pyrazinamide 25 mg/kg, Ethambutol 15–20 mg/kg × 2 months then Rifampin + Isoniazid × 4 months (6 months total); add Pyridoxine 25–50 mg daily<br>• Pregnancy: RIPE with Pyridoxine is used (Pyrazinamide is recommended by WHO but omitted by US guidance unless HIV or resistance; without it, give 9 months of Isoniazid + Rifampin + Ethambutol); avoid Streptomycin (fetal ototoxicity); breastfeeding is safe<br>• Renal/dialysis: Isoniazid and Rifampin no adjustment; Pyrazinamide and Ethambutol 3 times weekly (25 mg/kg and 15–25 mg/kg after dialysis) if CrCl <30<br>• Liver: hold if ALT ≥5× ULN (≥3× with symptoms) or bilirubin rise; re-challenge Rifampin first then Isoniazid then Pyrazinamide after ALT <2× ULN; alternatives: Levofloxacin, Ethambutol, Amikacin<br>• Interactions: Rifampin induces CYP3A4: Warfarin, DOACs, Tacrolimus, antiretrovirals (use Rifabutin or adjust Dolutegravir to 50 mg BID), azoles (Voriconazole: use Amphotericin), oral contraceptives, steroids<br>• HIV: start ART 2 weeks after TB therapy if CD4 <50 (and within 8 weeks if ≥50); Dolutegravir 50 mg BID with Rifampin<br>• Counseling: take on an empty stomach, adherence for the full course, orange body fluids (Rifampin), vision changes (Ethambutol: stop and call), jaundice/dark urine/nausea (stop and call), gout (Pyrazinamide), neuropathy (Isoniazid: Pyridoxine), avoid alcohol; hormonal contraceptives fail<br>• Monitoring: LFTs baseline, 2, 4, 8 weeks (monthly if risk), visual acuity and color vision monthly (Ethambutol), uric acid, sputum at 2 months, weight |

## Case 2: Isoniazid Hepatotoxicity (Difficult Patient) ★ Repeated case, also in 2021, 2018–2019

**Case details:** Difficult patient on Isoniazid with a significant LFT increase. Missed the appointment and came to collect the medication.

| Question | Answer |
|---|---|
| **Q1. Your recommendation and counseling?** | • Approach: stay calm and empathetic; do not blame the patient for the missed visit; use open questions and teach-back<br>• Do not dispense Isoniazid yet: check symptoms (nausea, vomiting, anorexia, dark urine, jaundice, RUQ pain) and review LFTs today<br>• Rule out other causes: alcohol, Acetaminophen and OTCs, herbals, diet, viral hepatitis<br>• ALT ≥3× ULN with symptoms, or ≥5× without: stop Isoniazid (and all TB drugs); inform the physician and monitor LFTs weekly<br>• ALT <3× ULN and asymptomatic: may continue with close follow-up<br>• Counsel: warning signs and when to call or go to the ED, avoid alcohol, limit Acetaminophen to ≤2 g/day, take Pyridoxine, reschedule follow-up, explain that most injury reverses when the drug is stopped<br>• Document and report the adverse drug reaction |
| **Also be ready for (exam add-ons)** | • Goals: cure, prevent transmission, prevent resistance and relapse, minimize hepatotoxicity<br>• Non-pharm: airborne isolation until 3 negative smears (or 2 weeks effective therapy and clinical improvement), contact tracing, DOT (directly observed therapy), nutrition, stop alcohol, HIV test<br>• First-line: RIPE: Rifampin 10 mg/kg (600 mg), Isoniazid 5 mg/kg (300 mg), Pyrazinamide 25 mg/kg, Ethambutol 15–20 mg/kg × 2 months then Rifampin + Isoniazid × 4 months (6 months total); add Pyridoxine 25–50 mg daily<br>• Pregnancy: RIPE with Pyridoxine is used (Pyrazinamide is recommended by WHO but omitted by US guidance unless HIV or resistance; without it, give 9 months of Isoniazid + Rifampin + Ethambutol); avoid Streptomycin (fetal ototoxicity); breastfeeding is safe<br>• Renal/dialysis: Isoniazid and Rifampin no adjustment; Pyrazinamide and Ethambutol 3 times weekly (25 mg/kg and 15–25 mg/kg after dialysis) if CrCl <30 |

| Question | Answer |
|---|---|
| | • Liver: hold if ALT ≥5× ULN (≥3× with symptoms) or bilirubin rise; re-challenge Rifampin first then Isoniazid then Pyrazinamide after ALT <2× ULN; alternatives: Levofloxacin, Ethambutol, Amikacin<br>• Interactions: Rifampin induces CYP3A4: Warfarin, DOACs, Tacrolimus, antiretrovirals (use Rifabutin or adjust Dolutegravir to 50 mg BID), azoles (Voriconazole: use Amphotericin), oral contraceptives, steroids<br>• HIV: start ART 2 weeks after TB therapy if CD4 <50 (and within 8 weeks if ≥50); Dolutegravir 50 mg BID with Rifampin<br>• Counseling: take on an empty stomach, adherence for the full course, orange body fluids (Rifampin), vision changes (Ethambutol: stop and call), jaundice/dark urine/nausea (stop and call), gout (Pyrazinamide), neuropathy (Isoniazid: Pyridoxine), avoid alcohol; hormonal contraceptives fail<br>• Monitoring: LFTs baseline, 2, 4, 8 weeks (monthly if risk), visual acuity and color vision monthly (Ethambutol), uric acid, sputum at 2 months, weight |

## Case 3: Meningococcal Encephalitis and Endocarditis (with Gentamicin TDM) ★ Repeated case, also in 2022–2023, 2018–2019

**Case details:** 53-year-old man with suspected meningococcal encephalitis, lumbar puncture pending. Headache, Brudzinski sign, leukocytosis, fever, seizures. Has a prosthetic valve.

| Question | Answer |
|---|---|
| **Q1. What are the signs and symptoms related to meningitis in this patient?** | • Fever and chills<br>• Severe headache<br>• Nuchal rigidity; positive Brudzinski and Kernig signs<br>• Altered mental status, seizures<br>• Photophobia, nausea and vomiting<br>• Petechiae or purpura (N. meningitidis)<br>• Leukocytosis |
| **Q2. What are the 5 suspected pathogens?** | • Streptococcus pneumoniae<br>• Neisseria meningitidis<br>• Listeria monocytogenes (age >50)<br>• Streptococcus agalactiae (group B)<br>• Haemophilus influenzae<br>• Encephalitis: HSV (cover with Acyclovir) |
| **Q3. What empiric antibiotics will you start (dose and frequency)?** | • Ceftriaxone 2 g IV q12h<br>• Vancomycin 15–20 mg/kg IV q8–12h (AUC-guided); covers resistant pneumococcus<br>• Ampicillin 2 g IV q4h (Listeria; age >50)<br>• Acyclovir 10 mg/kg IV q8h (encephalitis; hydrate)<br>• Dexamethasone 10 mg IV q6h (see Q4)<br>• Start without waiting for the LP; get blood cultures first; CT before LP only if indicated<br>• Monitor: neuro checks, temperature, CSF cultures/PCR, SCr (Acyclovir and Vancomycin), Vancomycin AUC |
| **Q4. What agent other than antibiotics should you give? For how long?** | • Dexamethasone 0.15 mg/kg (10 mg) IV q6h for 2–4 days<br>• Timing: 10–20 minutes before or with the first antibiotic dose<br>• Stop if the pathogen is not S. pneumoniae (or H. influenzae in children)<br>• Monitor: glucose, GI bleeding |
| **Q5. The team wants chemoprophylaxis. What options are available?** | • For close contacts of N. meningitidis: household, day care, exposure to oral secretions<br>• Rifampin 600 mg PO q12h × 4 doses<br>• Ciprofloxacin 500 mg PO × 1 dose<br>• Ceftriaxone 250 mg IM × 1 dose<br>• Azithromycin 500 mg PO × 1 dose<br>• Counsel: Rifampin turns urine and secretions orange, reduces oral contraceptive effect |
| **Q6. Later endocarditis, culture MSSA. Options for MSSA, and for a prosthetic valve?** | • Native valve MSSA: Nafcillin or Oxacillin 2 g IV q4h, or Cefazolin 2 g IV q8h × 4–6 weeks<br>• Prosthetic valve: the same β-lactam for ≥6 weeks plus Rifampin 300 mg PO/IV q8h (≥6 weeks) plus Gentamicin for the first 2 weeks<br>• Penicillin allergy: Cefazolin (if non-severe) or Vancomycin<br>• MRSA alternative: Vancomycin or Daptomycin 8–10 mg/kg<br>• Surgery consultation for prosthetic valve endocarditis<br>• Monitor: blood cultures daily until negative, echo, CRP, SCr, LFTs (Rifampin), drug interactions |
| **Q7. The team decided to start Gentamicin. What dosing do you recommend (normal renal function)?** | • Synergy dosing: 3 mg/kg/day IV in 2–3 divided doses (eg 1 mg/kg q8h) for the first 2 weeks<br>• Weight: ideal body weight (adjusted if obese)<br>• Not extended-interval dosing (this is synergy, not killing by peak)<br>• Monitor: SCr and urine output 2–3 times/week, levels, K/Mg, hearing and balance |
| **Q8. What is your Gentamicin trough level target?** | • Trough <1 mg/L (endocarditis synergy); peak 3–4 mg/L about 30 minutes after infusion<br>• General target: trough <2 mg/L<br>• Timing: at steady state (after the 3rd–4th dose), trough drawn within 30 minutes before the next dose |
| **Also be ready for (exam add-ons)** | • Goals: rapid bactericidal therapy (within 1 h), reduce mortality and neurologic sequelae, prevent spread |

| Question | Answer |
|---|---|
| | • Non-pharm: droplet isolation for 24 h (N. meningitidis), ICU if needed, CT before LP only if indicated, contact prophylaxis, hearing screen, vaccination<br>• First-line: age >50 or immunocompromised: Ceftriaxone 2 g q12h + Vancomycin (AUC/trough 15–20) + Ampicillin 2 g q4h (Listeria) + Dexamethasone 0.15 mg/kg q6h × 4 days (before or with the first antibiotic dose); add Acyclovir 10 mg/kg q8h if encephalitis suspected; S. pneumoniae MIC <0.1: Penicillin G or Ampicillin; N. meningitidis: Ceftriaxone/Penicillin G × 7 days; Listeria: Ampicillin 2–3 weeks<br>• Pregnancy: Ceftriaxone, Ampicillin and Vancomycin are safe; avoid aminoglycosides (ototoxic); Dexamethasone acceptable; Acyclovir acceptable; prophylaxis: Ceftriaxone 250 mg IM (Ciprofloxacin and Rifampin avoided)<br>• Renal/dialysis: Acyclovir adjust (CrCl <50: q12h) and hydrate; Vancomycin AUC; Ampicillin reduce interval; Ceftriaxone no change<br>• Liver: Ceftriaxone ≤2 g/day in combined hepatic and renal failure; Rifampin hepatotoxic<br>• Penicillin allergy: Chloramphenicol or Meropenem (low cross-reactivity) or desensitize for Listeria (TMP-SMX 15–20 mg/kg/day)<br>• Chemoprophylaxis (meningococcus, close contacts within 24 h): Rifampin 600 mg q12h × 2 days, Ciprofloxacin 500 mg once, or Ceftriaxone 250 mg IM once<br>• Counseling: vaccination (MenACWY, MenB, pneumococcal), hearing and neurologic follow-up, complete the course, steroids side effects<br>• Monitoring: neuro checks, temperature, repeat LP at 48 h if no response, CSF cultures, SCr, Vancomycin level, seizures, glucose |

## Case 4: Community-Acquired Pneumonia (Anemia, Ofloxacin Anaphylaxis, Gentamicin TDM) ★ Repeated case, also in 2018–2019

**Case details:** 60-year-old with fever, cough and chest pain, admitted to the medical ward. Chest X-ray: multilobar infiltrate. Anemia on Ferrous sulfate; anaphylaxis to Ofloxacin. T 39 °C, WBC 14.

| Question | Answer |
|---|---|
| **Q1. Discuss the diagnosis and classify the pneumonia.** | • CAP requiring hospitalization (PSI >90 or clinical judgment)<br>• Severity (IDSA/ATS): 1 major (shock needing pressors, mechanical ventilation) or ≥3 minor (RR >30, PaO2/FiO2 ≤250, multilobar infiltrates, confusion, BUN ≥20, WBC <4000, platelets <100,000, temp <36, hypotension)<br>• This patient: multilobar infiltrate = 1 minor → non-severe unless other criteria are met |
| **Q2. What are the common microorganisms?** | • S. pneumoniae (most common)<br>• H. influenzae<br>• Atypicals: Mycoplasma, Chlamydophila, Legionella<br>• S. aureus, Enterobacterales, Moraxella catarrhalis, respiratory viruses |
| **Q3. What to start?** | • Non-severe inpatient: β-lactam + atypical coverage<br>  ◦ Ampicillin-sulbactam 3 g IV q6h or Ceftriaxone 1–2 g IV q24h<br>  ◦ plus Azithromycin 500 mg IV/PO daily or Doxycycline 100 mg PO/IV q12h<br>• Avoid respiratory fluoroquinolones: Ofloxacin anaphylaxis → class cross-reactivity<br>• Duration: at least 5 days and clinical stability<br>• Monitor: temperature, respiratory rate, SpO2, WBC, chest X-ray if not improving, cultures<br>• Non-pharm: hydration, oxygen, early mobilization, vaccines before discharge |
| **Q4. Started on Azithromycin and has QT prolongation. Your action?** | • Stop Azithromycin: switch to Doxycycline 100 mg q12h (no QT effect)<br>• ECG/QTc: repeat; correct K >4 and Mg >2<br>• Review other QT drugs; avoid fluoroquinolones (also contraindicated here) |
| **Q5. What to give to cover atypical bacteria?** | • Doxycycline 100 mg PO/IV q12h<br>• Macrolide (Azithromycin or Clarithromycin) only if QT is acceptable<br>• Not a fluoroquinolone (anaphylaxis)<br>• Newer non-quinolone option: Omadacycline 200 mg IV on day 1 then 100 mg daily (separate oral doses from iron); Lefamulin is avoided with QT prolongation |
| **Q6. Doxycycline drug-drug interactions and counseling?** | • Ferrous sulfate and other cations: iron, calcium, magnesium, aluminum, zinc, antacids, dairy → chelation lowers absorption<br>  ◦ Separate: Doxycycline 2 hours before or 3–4 hours after iron/antacids<br>• Warfarin: INR rises, check INR<br>• Counsel: full glass of water, sit upright 30 min (esophagitis), sun protection, take with food if GI upset (not with dairy), finish the course |
| **Q7. Culture sensitive only to Gentamicin: which dosing strategy, calculate the dose, TDM?** | • Strategy: extended-interval (once daily) 5–7 mg/kg IV q24h using ideal (or adjusted) body weight<br>• Example: IBW 70 kg × 7 mg/kg = 490 mg q24h (adjust interval if CrCl <40–60 mL/min)<br>• Conventional alternative: 1.5–2 mg/kg q8h<br>• TDM: level 6–14 hours after the first dose plotted on the Hartford nomogram; or peak/trough<br>• Targets: peak 4–10 mg/L (conventional; higher for extended-interval), trough <1–2 mg/L (undetectable for extended-interval)<br>• Trough timing: drawn within 30 minutes before the next dose, after the 3rd–4th dose<br>• Caution: poor lung penetration and inactivation at low pH; use as part of combination therapy |

| Question | Answer |
|---|---|
| **Q8. Side effects of Gentamicin, and what to monitor?** | • Nephrotoxicity: non-oliguric, SCr rises after 5–10 days; K and Mg wasting<br>• Ototoxicity: cochlear and vestibular, often irreversible<br>• Neuromuscular blockade<br>• Monitor: SCr, BUN and urine output baseline and 2–3 times weekly; levels; K, Mg; hearing and balance; limit to 7–10 days<br>• Prevention: hydrate, avoid other nephrotoxins (Vancomycin, loop diuretics, contrast), extended-interval dosing |
| **Also be ready for (exam add-ons)** | • Goals: cure the infection, prevent complications, minimize resistance and QT/drug interactions, 5-day course<br>• Non-pharm: severity assessment (CURB-65, IDSA/ATS), oxygen, hydration, rest, smoking cessation, vaccines (pneumococcal, influenza, COVID-19, RSV), follow-up chest X-ray only if indicated<br>• First-line: inpatient non-severe: Ceftriaxone 1–2 g daily + Azithromycin 500 mg daily (or Doxycycline 100 mg BID); with penicillin allergy: respiratory fluoroquinolone (Levofloxacin 750 mg daily or Moxifloxacin 400 mg) or Doxycycline; severe: β-lactam + macrolide; add MRSA/Pseudomonas coverage only with risk factors (Vancomycin or Linezolid; Cefepime/Pip-tazo)<br>• Pregnancy: Amoxicillin, Ceftriaxone, Azithromycin safe; avoid Doxycycline and fluoroquinolones; Pneumococcal/influenza vaccines safe<br>• Renal/dialysis: Levofloxacin 750 mg q48h if CrCl 20–49; Gentamicin extended-interval by CrCl with levels; Azithromycin no change<br>• Liver: Azithromycin/Moxifloxacin caution in severe disease<br>• Interactions: Azithromycin/quinolones QT prolongation (stop and use Doxycycline); Doxycycline with ferrous sulfate/antacids (separate by 2–3 h); Levofloxacin tendon risk with steroids; ofloxacin anaphylaxis: avoid all fluoroquinolones<br>• Counseling: complete the course, Doxycycline: sun protection, take upright with water, avoid dairy/iron close to the dose; Azithromycin: palpitations; Levofloxacin: tendon pain, dysglycemia<br>• Gentamicin TDM: extended-interval 5–7 mg/kg q24h (trough <1 mg/L), conventional peak 5–10, trough <2; SCr, hearing/vestibular function |

## Case 5: Bacterial Meningitis Progressing to Hypertensive Emergency ★ Repeated case, also in 2022–2023, 2018–2019

**Case details:** 62-year-old with DM and HTN, severe headache (worst of his life, not relieved by Ibuprofen), febrile, elevated WBC; CSF not yet done.

| Question | Answer |
|---|---|
| **Q1. What further investigations?** | • Lumbar puncture: opening pressure, cell count and differential, glucose (with simultaneous serum glucose), protein, Gram stain, culture, PCR<br>• Blood cultures before antibiotics<br>• CT head before LP only if: immunocompromised, focal deficit, seizure, papilledema, altered consciousness<br>• Do not delay antibiotics and Dexamethasone for imaging<br>• Others: CBC, CRP, BMP, glucose |
| **Q2. They will show the results: what will you look at?** | • WBC >1000/µL, mostly neutrophils<br>• Glucose <50 mg/dL (CSF:serum ≤0.4)<br>• Protein >150 mg/dL<br>• Opening pressure elevated; appearance turbid<br>• Gram stain (positive in 60–90%) and culture |
| **Q3. Suspected organism in this age group?** | • S. pneumoniae<br>• N. meningitidis<br>• Listeria monocytogenes (age >50)<br>• Less common: S. agalactiae, H. influenzae, gram-negative bacilli |
| **Q4. Suggested empiric therapy?** | • Ceftriaxone 2 g IV q12h<br>• Vancomycin 15–20 mg/kg IV q8–12h<br>• Ampicillin 2 g IV q4h<br>• Dexamethasone 10 mg IV q6h for 2–4 days, with or before the first antibiotic dose<br>• Monitor: neuro status, temperature, glucose (steroid), SCr, Vancomycin AUC, repeat CSF at 48 h if no improvement |
| **Q5. Culture revealed S. pneumoniae with MIC <0.1: your action and duration?** | • De-escalate: Penicillin G 4 million units IV q4h or Ampicillin 2 g IV q4h (alternative Ceftriaxone 2 g q12h)<br>• Stop Vancomycin (and Ampicillin if not Listeria) once susceptibility is confirmed<br>• Continue Dexamethasone to complete 4 days<br>• Duration: 10–14 days |
| **Q6. After 10 days he returned with headache after taking Ibuprofen, and a visual problem. BP 220/120. Your assessment?** | • Hypertensive emergency (BP >180/120 with target-organ damage: visual change, headache)<br>• Contributor: NSAIDs cause fluid and sodium retention and raise BP<br>• Also exclude meningitis complications (hydrocephalus, subdural, venous thrombosis) with imaging<br>• Stop Ibuprofen; admit to monitored unit |

| Question | Answer |
|---|---|
| **Q7. What is your goal?** | • First hour: lower BP by no more than 25% (MAP)<br>• Next 2–6 hours: to about 160/100–110<br>• Then to normal over 24–48 hours<br>• Avoid overshoot (cerebral hypoperfusion) |
| **Q8. How will you achieve the target?** | • Labetalol 20 mg IV over 2 min, then 20–80 mg q10 min (max 300 mg), or infusion 0.5–2 mg/min<br>• Alternative: Nicardipine 5 mg/h IV, titrate by 2.5 mg/h every 5–15 min (max 15 mg/h); Clevidipine 1–2 mg/h<br>• Monitor: BP every 5–15 min (arterial line), HR, neuro and visual status, fundoscopy, SCr, troponin, ECG<br>• Transition to oral agents once stable; avoid NSAIDs |
| **Also be ready for (exam add-ons)** | • Goals: rapid bactericidal therapy (within 1 h), reduce mortality and neurologic sequelae, prevent spread<br>• Non-pharm: droplet isolation for 24 h (N. meningitidis), ICU if needed, CT before LP only if indicated, contact prophylaxis, hearing screen, vaccination<br>• First-line: age >50 or immunocompromised: Ceftriaxone 2 g q12h + Vancomycin (AUC/trough 15–20) + Ampicillin 2 g q4h (Listeria) + Dexamethasone 0.15 mg/kg q6h × 4 days (before or with the first antibiotic dose); add Acyclovir 10 mg/kg q8h if encephalitis suspected; S. pneumoniae MIC <0.1: Penicillin G or Ampicillin; N. meningitidis: Ceftriaxone/Penicillin G × 7 days; Listeria: Ampicillin 2–3 weeks<br>• Pregnancy: Ceftriaxone, Ampicillin and Vancomycin are safe; avoid aminoglycosides (ototoxic); Dexamethasone acceptable; Acyclovir acceptable; prophylaxis: Ceftriaxone 250 mg IM (Ciprofloxacin and Rifampin avoided)<br>• Renal/dialysis: Acyclovir adjust (CrCl <50: q12h) and hydrate; Vancomycin AUC; Ampicillin reduce interval; Ceftriaxone no change<br>• Liver: Ceftriaxone ≤2 g/day in combined hepatic and renal failure; Rifampin hepatotoxic<br>• Penicillin allergy: Chloramphenicol or Meropenem (low cross-reactivity) or desensitize for Listeria (TMP-SMX 15–20 mg/kg/day)<br>• Chemoprophylaxis (meningococcus, close contacts within 24 h): Rifampin 600 mg q12h × 2 days, Ciprofloxacin 500 mg once, or Ceftriaxone 250 mg IM once<br>• Counseling: vaccination (MenACWY, MenB, pneumococcal), hearing and neurologic follow-up, complete the course, steroids side effects<br>• Monitoring: neuro checks, temperature, repeat LP at 48 h if no response, CSF cultures, SCr, Vancomycin level, seizures, glucose |

## Case 6: Bacteremia on Hemodialysis (MRSA and ESBL, Vancomycin TDM)

**Case details:** Hemodialysis patient with bacteremia; echocardiogram negative for infective endocarditis.

| Question | Answer |
|---|---|
| **Q1. What antibiotic will you give?** | • Empiric broad coverage: gram-positive (including MRSA) and gram-negative<br>  ◦ Vancomycin IV<br>  ◦ plus an antipseudomonal β-lactam: Cefepime 1 g IV q24h after HD, or Piperacillin-tazobactam 2.25 g q8h + 0.75 g after HD<br>• Move to a carbapenem if ESBL risk or ESBL isolated<br>• Source control: remove or exchange the dialysis catheter<br>• Blood cultures ×2 before the first dose |
| **Q2. What is the starting dose of Vancomycin (LD and maintenance)?** | • Loading dose: 25 mg/kg actual body weight (max ~3 g), given after HD<br>• Maintenance: 5–10 mg/kg after each HD session<br>• Adjust to pre-HD level / AUC |
| **Q3. When will you draw the trough?** | • Before the next dialysis session (pre-HD level)<br>• Targets: AUC 400–600 mg·h/L preferred; pre-HD 15–20 mg/L is the common practical target |
| **Q4. Culture positive for MRSA and ESBL: what will you change?** | • Meropenem 500 mg IV q24h after HD (renal dose; 1 g q24h in severe infection) plus Vancomycin<br>• Alternative for ESBL: Ertapenem 500 mg IV q24h (CrCl ≤30) if not critically ill and not hypoalbuminemic<br>• Stop Cefepime or Piperacillin-tazobactam (not reliable for ESBL bacteremia)<br>• Monitor: Vancomycin levels, repeat blood cultures 48–72 h, fever, WBC, catheter site, seizure risk (carbapenem, ESRD)<br>• Newer option for MRSA in hemodialysis: Dalbavancin 1500 mg IV, repeated after 1 week (long-acting lipoglycopeptide; no dose adjustment on HD) |
| **Q5. From which culture will you count the duration?** | • MRSA: count from the first negative blood culture; at least 14 days if uncomplicated (echo negative, defervescence within 72 h, no prosthetic material, repeat cultures negative)<br>• Complicated MRSA bacteremia: 4–6 weeks<br>• ESBL bacteremia with source control and response: about 7 days |
| **Also be ready for (exam add-ons)** | • Goals: clear bacteremia, rule out endocarditis, source control, avoid vancomycin toxicity<br>• Non-pharm: remove/replace the HD catheter or line, echocardiography, repeat blood cultures, vascular access hygiene<br>• First-line: broad empiric (Vancomycin + Cefepime or Meropenem), then MRSA: Vancomycin (load 25 mg/kg, redose 500–1000 mg after HD, trough 15–20); ESBL: Meropenem 1 g q24h (HD after session); duration 14 days from the first negative culture (uncomplicated), 4–6 weeks if endocarditis/osteomyelitis<br>• Pregnancy: Vancomycin and Meropenem acceptable; avoid aminoglycosides, fluoroquinolones |

| Question | Answer |
|---|---|
| | • Renal/dialysis: dose after dialysis, Vancomycin pre-dialysis level, Cefazolin 2 g after HD alternative for MSSA<br>• Liver: no major adjustment<br>• Counseling: dialysis hygiene, report fever/chills after HD, finish therapy<br>• Monitoring: blood cultures every 24–48 h until negative, pre-HD Vancomycin level, SCr, temperature, CBC |

# 2021

## Case 1: GERD, Type 1 DM and Latent TB on Isoniazid (Hepatotoxicity) ★ Repeated case, also in 2022–2023, 2018–2019

**Case details:** Patient with GERD, type 1 DM and latent TB, on Isoniazid for 5 months. Presented with hepatotoxicity.

| Question | Answer |
|---|---|
| **Q1. How to manage and counsel the patient?** | • Assess: symptoms (nausea, vomiting, jaundice, dark urine, RUQ pain), ALT/AST, bilirubin, ALP, INR<br>• Exclude other causes: alcohol, Acetaminophen, herbals, viral hepatitis, other drugs<br>• Stop Isoniazid if ALT ≥3× ULN with symptoms or ≥5× ULN without symptoms; do not rechallenge Isoniazid after a clear Isoniazid hepatitis<br>• Later options: after normalization, the physician may complete latent TB therapy with Rifampin 600 mg daily to finish 4 months (4R), watching interactions<br>• Diabetes: Isoniazid can disturb glucose control, so check glucose more often; Rifampin interacts with oral hypoglycemics, not insulin<br>• GERD: aluminum/magnesium antacids reduce Isoniazid absorption → separate by ≥1 hour; PPIs are fine<br>• Counsel: warning signs and when to seek care, avoid alcohol, limit Acetaminophen, Pyridoxine 25–50 mg daily, explain most injury is reversible, keep follow-up<br>• Monitor: LFTs weekly until normal, then every 2–4 weeks; clinical review monthly<br>• Report the adverse drug reaction<br>• Newer short latent TB regimens (3HP: Rifapentine + Isoniazid weekly × 12; 3HR) contain Isoniazid, so avoid them after Isoniazid hepatitis; 4R is the right option |
| **Also be ready for (exam add-ons)** | • Goals: cure, prevent transmission, prevent resistance and relapse, minimize hepatotoxicity<br>• Non-pharm: airborne isolation until 3 negative smears (or 2 weeks effective therapy and clinical improvement), contact tracing, DOT (directly observed therapy), nutrition, stop alcohol, HIV test<br>• First-line: RIPE: Rifampin 10 mg/kg (600 mg), Isoniazid 5 mg/kg (300 mg), Pyrazinamide 25 mg/kg, Ethambutol 15–20 mg/kg × 2 months then Rifampin + Isoniazid × 4 months (6 months total); add Pyridoxine 25–50 mg daily<br>• Pregnancy: RIPE with Pyridoxine is used (Pyrazinamide is recommended by WHO but omitted by US guidance unless HIV or resistance; without it, give 9 months of Isoniazid + Rifampin + Ethambutol); avoid Streptomycin (fetal ototoxicity); breastfeeding is safe<br>• Renal/dialysis: Isoniazid and Rifampin no adjustment; Pyrazinamide and Ethambutol 3 times weekly (25 mg/kg and 15–25 mg/kg after dialysis) if CrCl <30<br>• Liver: hold if ALT ≥5× ULN (≥3× with symptoms) or bilirubin rise; re-challenge Rifampin first then Isoniazid then Pyrazinamide after ALT <2× ULN; alternatives: Levofloxacin, Ethambutol, Amikacin<br>• Interactions: Rifampin induces CYP3A4: Warfarin, DOACs, Tacrolimus, antiretrovirals (use Rifabutin or adjust Dolutegravir to 50 mg BID), azoles (Voriconazole: use Amphotericin), oral contraceptives, steroids<br>• HIV: start ART 2 weeks after TB therapy if CD4 <50 (and within 8 weeks if ≥50); Dolutegravir 50 mg BID with Rifampin<br>• Counseling: take on an empty stomach, adherence for the full course, orange body fluids (Rifampin), vision changes (Ethambutol: stop and call), jaundice/dark urine/nausea (stop and call), gout (Pyrazinamide), neuropathy (Isoniazid: Pyridoxine), avoid alcohol; hormonal contraceptives fail<br>• Monitoring: LFTs baseline, 2, 4, 8 weeks (monthly if risk), visual acuity and color vision monthly (Ethambutol), uric acid, sputum at 2 months, weight |

# 2020

## Cases 1 and 3: TB with HIV (Repeated Station) ★ Repeated case, also in 2020

**Case details:** TB with HIV, CD4 82–85 cells/mm$^3$. Station repeated across the exam.

| Question | Answer |
|---|---|
| **Q1. What to initiate first, and starting agents?** | • Start TB therapy first: Isoniazid, Rifampin, Pyrazinamide, Ethambutol + Pyridoxine daily (Rifabutin in place of Rifampin if the ART uses a protease inhibitor)<br>• Then start ART (see below)<br>• TB therapy by directly observed therapy |

| Question | Answer |
|---|---|
| **Q2. When to start ART (CD4 82)? Why not start together?** | • CD4 <50: ART within 2 weeks<br>• CD4 ≥50 (82–85): ART within 8 weeks of TB therapy<br>• Why not together: **IRIS**, overlapping toxicities (hepatitis, rash), pill burden and adherence, drug interactions<br>• Exception: TB meningitis needs careful timing and steroids because of higher risk |
| **Q3. What interactions are there?** | • Rifampin induces CYP3A4/UGT: lowers protease inhibitors, rilpivirine, doravirine, Elvitegravir/cobicistat, Bictegravir, TAF<br>• Dolutegravir: increase to 50 mg BID with Rifampin<br>• Efavirenz: acceptable, 600 mg daily<br>• Rifabutin alternative: 300 mg daily (150 mg daily with ritonavir-boosted PI); avoid with cobicistat and with Elvitegravir/Bictegravir<br>• Monitor: HIV RNA, ART and Rifabutin levels where available |
| **Q4. Give an example of an ART regimen.** | • Tenofovir DF 300 mg + Emtricitabine 200 mg daily + Dolutegravir 50 mg BID while on Rifampin (daily again 2 weeks after Rifampin stops)<br>• Alternative: Tenofovir DF/Emtricitabine + Efavirenz 600 mg daily<br>• Avoid TAF with Rifampin |
| **Q5. Duration of TB therapy?** | • 6 months (2 months RIPE + 4 months Isoniazid/Rifamycin)<br>• 9 months if cavitary disease with positive culture at 2 months<br>• Monitor: sputum smear/culture monthly until negative, LFTs, visual acuity, HIV RNA, CD4 |
| **Q6. Criteria to remove airborne isolation (and when to stop the negative-pressure room)?** | • Airborne isolation: negative-pressure room, N95 respirator for staff<br>• Discontinue when: ≥2 weeks of effective therapy, clinical improvement (cough and fever resolved), and 3 consecutive negative sputum AFB smears collected ≥8 hours apart (one early morning)<br>• HIV or drug-resistant TB: stricter, follow infection control |
| **Also be ready for (exam add-ons)** | • Goals: cure, prevent transmission, prevent resistance and relapse, minimize hepatotoxicity<br>• Non-pharm: airborne isolation until 3 negative smears (or 2 weeks effective therapy and clinical improvement), contact tracing, DOT (directly observed therapy), nutrition, stop alcohol, HIV test<br>• First-line: RIPE: Rifampin 10 mg/kg (600 mg), Isoniazid 5 mg/kg (300 mg), Pyrazinamide 25 mg/kg, Ethambutol 15–20 mg/kg × 2 months then Rifampin + Isoniazid × 4 months (6 months total); add Pyridoxine 25–50 mg daily<br>• Pregnancy: RIPE with Pyridoxine is used (Pyrazinamide is recommended by WHO but omitted by US guidance unless HIV or resistance; without it, give 9 months of Isoniazid + Rifampin + Ethambutol); avoid Streptomycin (fetal ototoxicity); breastfeeding is safe<br>• Renal/dialysis: Isoniazid and Rifampin no adjustment; Pyrazinamide and Ethambutol 3 times weekly (25 mg/kg and 15–25 mg/kg after dialysis) if CrCl <30<br>• Liver: hold if ALT ≥5× ULN (≥3× with symptoms) or bilirubin rise; re-challenge Rifampin first then Isoniazid then Pyrazinamide after ALT <2× ULN; alternatives: Levofloxacin, Ethambutol, Amikacin<br>• Interactions: Rifampin induces CYP3A4: Warfarin, DOACs, Tacrolimus, antiretrovirals (use Rifabutin or adjust Dolutegravir to 50 mg BID), azoles (Voriconazole: use Amphotericin), oral contraceptives, steroids<br>• HIV: start ART 2 weeks after TB therapy if CD4 <50 (and within 8 weeks if ≥50); Dolutegravir 50 mg BID with Rifampin<br>• Counseling: take on an empty stomach, adherence for the full course, orange body fluids (Rifampin), vision changes (Ethambutol: stop and call), jaundice/dark urine/nausea (stop and call), gout (Pyrazinamide), neuropathy (Isoniazid: Pyridoxine), avoid alcohol; hormonal contraceptives fail<br>• Monitoring: LFTs baseline, 2, 4, 8 weeks (monthly if risk), visual acuity and color vision monthly (Ethambutol), uric acid, sputum at 2 months, weight |

## Case 2: HIV-TB (Detailed) ★ Repeated case, also in 2020

**Case details:** HIV patient diagnosed with TB. Team suggests Elvitegravir/cobicistat/Emtricitabine/Tenofovir. Concern about TB and HIV drug interactions.

| Question | Answer |
|---|---|
| **Q1. What is your recommended regimen for both HIV and TB?** | • TB: Isoniazid + Rifampin + Pyrazinamide + Ethambutol + Pyridoxine<br>• HIV: do not use Elvitegravir/cobicistat (levels fall with Rifampin)<br>  ◦ Tenofovir DF + Emtricitabine + Dolutegravir 50 mg BID<br>  ◦ or Tenofovir DF + Emtricitabine + Efavirenz 600 mg daily |
| **Q2. Which agents as ART, and the dose adjustment?** | • Dolutegravir: 50 mg q12h with Rifampin<br>• Efavirenz: 600 mg daily, no adjustment<br>• Raltegravir: 800 mg BID<br>• If a PI is needed: use Rifabutin 150 mg daily (with ritonavir) instead of Rifampin<br>• Monitor: HIV RNA, LFTs, mood and CNS effects (Efavirenz) |
| **Q3. Which Tenofovir will you use, and why (TDF vs TAF)?** | • Tenofovir DF: Rifampin lowers TAF concentrations, so TAF is not recommended with Rifampin<br>• Monitor TDF: SCr, phosphate, urine protein |
| **Q4. Which will you treat first (TB or HIV), and why?** | • TB first, ART started within 2–8 weeks<br>• Why: lower IRIS risk, fewer overlapping toxicities, better adherence<br>• Exception: CD4 <50: ART within 2 weeks |

| Question | Answer |
|---|---|
| **Q5. How long will you treat TB?** | • 6 months, extended to 9 months if cavitary with positive culture at 2 months |
| **Q6. When will you start ART (based on CD4)?** | • CD4 <50: within 2 weeks<br>• CD4 ≥50: within 8 weeks |
| **Also be ready for (exam add-ons)** | • Goals: cure, prevent transmission, prevent resistance and relapse, minimize hepatotoxicity<br>• Non-pharm: airborne isolation until 3 negative smears (or 2 weeks effective therapy and clinical improvement), contact tracing, DOT (directly observed therapy), nutrition, stop alcohol, HIV test<br>• First-line: RIPE: Rifampin 10 mg/kg (600 mg), Isoniazid 5 mg/kg (300 mg), Pyrazinamide 25 mg/kg, Ethambutol 15–20 mg/kg × 2 months then Rifampin + Isoniazid × 4 months (6 months total); add Pyridoxine 25–50 mg daily<br>• Pregnancy: RIPE with Pyridoxine is used (Pyrazinamide is recommended by WHO but omitted by US guidance unless HIV or resistance; without it, give 9 months of Isoniazid + Rifampin + Ethambutol); avoid Streptomycin (fetal ototoxicity); breastfeeding is safe<br>• Renal/dialysis: Isoniazid and Rifampin no adjustment; Pyrazinamide and Ethambutol 3 times weekly (25 mg/kg and 15–25 mg/kg after dialysis) if CrCl <30<br>• Liver: hold if ALT ≥5× ULN (≥3× with symptoms) or bilirubin rise; re-challenge Rifampin first then Isoniazid then Pyrazinamide after ALT <2× ULN; alternatives: Levofloxacin, Ethambutol, Amikacin<br>• Interactions: Rifampin induces CYP3A4: Warfarin, DOACs, Tacrolimus, antiretrovirals (use Rifabutin or adjust Dolutegravir to 50 mg BID), azoles (Voriconazole: use Amphotericin), oral contraceptives, steroids<br>• HIV: start ART 2 weeks after TB therapy if CD4 <50 (and within 8 weeks if ≥50); Dolutegravir 50 mg BID with Rifampin<br>• Counseling: take on an empty stomach, adherence for the full course, orange body fluids (Rifampin), vision changes (Ethambutol: stop and call), jaundice/dark urine/nausea (stop and call), gout (Pyrazinamide), neuropathy (Isoniazid: Pyridoxine), avoid alcohol; hormonal contraceptives fail<br>• Monitoring: LFTs baseline, 2, 4, 8 weeks (monthly if risk), visual acuity and color vision monthly (Ethambutol), uric acid, sputum at 2 months, weight |

# 2018–2019

## Case 1: Meningitis ★ Repeated case, also in 2022–2023

> **Case details:** Bacterial meningitis case.

| Question | Answer |
|---|---|
| **Q1. Signs and symptoms?** | • Fever and chills<br>• Headache and neck stiffness<br>• Photophobia, nausea and vomiting<br>• Altered mental status, seizures<br>• Kernig and Brudzinski signs<br>• Petechiae or purpura (meningococcus) |
| **Q2. Regimen with dosing and the bacterial coverage of each agent?** | • Ceftriaxone 2 g IV q12h: S. pneumoniae, N. meningitidis, H. influenzae, group B streptococcus<br>• Vancomycin 15–20 mg/kg IV q8–12h: penicillin-resistant S. pneumoniae<br>• Ampicillin 2 g IV q4h (age >50 or immunocompromised): Listeria, enterococci<br>• Acyclovir 10 mg/kg IV q8h: HSV (if encephalitis)<br>• Neonate (<1 month): Ampicillin + Cefotaxime or an aminoglycoside<br>• Post-neurosurgery/shunt: Vancomycin + Cefepime, Ceftazidime or Meropenem |
| **Q3. Common organisms?** | • <1 month: group B streptococcus, Listeria, E. coli<br>• 1–23 months: S. pneumoniae, N. meningitidis, H. influenzae<br>• 2–50 years: N. meningitidis, S. pneumoniae<br>• >50 years: S. pneumoniae, N. meningitidis, Listeria, gram-negative bacilli |
| **Q4. Monitoring?** | • Clinical: temperature, mental status/neuro checks, seizures<br>• Lab: WBC, CRP, SCr (Vancomycin/Acyclovir), Vancomycin AUC, glucose with Dexamethasone<br>• CSF: repeat at 48 hours if no improvement or resistant pneumococcus<br>• Complications: hearing test after recovery, imaging for hydrocephalus |
| **Q5. Resolution of symptoms?** | • Fever resolves, headache and neck stiffness improve<br>• Normal mental status, WBC normalizes<br>• CSF sterile and cell count/glucose improve<br>• Duration:<br>  ◦ N. meningitidis 7 days; H. influenzae 7 days<br>  ◦ S. pneumoniae 10–14 days<br>  ◦ Group B streptococcus 14–21 days; Listeria ≥21 days<br>  ◦ HSV 10–14 days (meningitis), 14–21 days (encephalitis) |

| Question | Answer |
|---|---|
| **Q6. What is the role of steroids?** | • Dexamethasone 0.15 mg/kg (10 mg) IV q6h for 2–4 days<br>• Benefit: lower mortality in pneumococcal meningitis in adults; less hearing loss with H. influenzae in children<br>• Timing: 10–20 minutes before or with the first antibiotic dose<br>• Stop if the pathogen is not S. pneumoniae or H. influenzae<br>• Caution: may lower Vancomycin CSF penetration |
| **Also be ready for (exam add-ons)** | • Goals: rapid bactericidal therapy (within 1 h), reduce mortality and neurologic sequelae, prevent spread<br>• Non-pharm: droplet isolation for 24 h (N. meningitidis), ICU if needed, CT before LP only if indicated, contact prophylaxis, hearing screen, vaccination<br>• First-line: age >50 or immunocompromised: Ceftriaxone 2 g q12h + Vancomycin (AUC/trough 15–20) + Ampicillin 2 g q4h (Listeria) + Dexamethasone 0.15 mg/kg q6h × 4 days (before or with the first antibiotic dose); add Acyclovir 10 mg/kg q8h if encephalitis suspected; S. pneumoniae MIC <0.1: Penicillin G or Ampicillin; N. meningitidis: Ceftriaxone/Penicillin G × 7 days; Listeria: Ampicillin 2–3 weeks<br>• Pregnancy: Ceftriaxone, Ampicillin and Vancomycin are safe; avoid aminoglycosides (ototoxic); Dexamethasone acceptable; Acyclovir acceptable; prophylaxis: Ceftriaxone 250 mg IM (Ciprofloxacin and Rifampin avoided)<br>• Renal/dialysis: Acyclovir adjust (CrCl <50: q12h) and hydrate; Vancomycin AUC; Ampicillin reduce interval; Ceftriaxone no change<br>• Liver: Ceftriaxone ≤2 g/day in combined hepatic and renal failure; Rifampin hepatotoxic<br>• Penicillin allergy: Chloramphenicol or Meropenem (low cross-reactivity) or desensitize for Listeria (TMP-SMX 15–20 mg/kg/day)<br>• Chemoprophylaxis (meningococcus, close contacts within 24 h): Rifampin 600 mg q12h × 2 days, Ciprofloxacin 500 mg once, or Ceftriaxone 250 mg IM once<br>• Counseling: vaccination (MenACWY, MenB, pneumococcal), hearing and neurologic follow-up, complete the course, steroids side effects<br>• Monitoring: neuro checks, temperature, repeat LP at 48 h if no response, CSF cultures, SCr, Vancomycin level, seizures, glucose |

## Case 2: Ventilator-Associated Pneumonia (VAP) ★ Repeated case, also in 2025

**Case details:** VAP case.

| Question | Answer |
|---|---|
| **Q1. Classifications?** | • VAP: pneumonia developing >48 hours after endotracheal intubation<br>• HAP: pneumonia ≥48 hours after admission in a non-ventilated patient<br>• CAP: acquired in the community or within 48 hours of admission<br>• HCAP was retired in 2016 |
| **Q2. Recommended regimen, dosing and duration for VAP?** | • Antipseudomonal β-lactam: Piperacillin-tazobactam 4.5 g IV q6h, Cefepime 2 g IV q8h, Meropenem 1 g IV q8h or Imipenem 500 mg q6h<br>• Add a second antipseudomonal agent if MDR risk (IV antibiotics within 90 days, ≥5 days in hospital, septic shock, ARDS, renal replacement) or unit resistance >10%: Gentamicin/Tobramycin or Ciprofloxacin 400 mg IV q8h / Levofloxacin 750 mg q24h<br>• Add MRSA coverage if MDR risk or unit MRSA >20% or unknown: Vancomycin (AUC-guided) or Linezolid 600 mg IV q12h<br>• Duration: 7 days<br>• Non-pharm (prevention): head of bed 30–45°, daily sedation interruption, oral care with toothbrushing (routine chlorhexidine is no longer advised: SHEA 2022), minimize ventilation days<br>• Monitor: temperature, WBC, oxygenation, secretions, cultures, SCr<br>• Ceftriaxone + Levofloxacin is a CAP regimen and does not reliably cover Pseudomonas or MRSA |
| **Q3. If cultures return, how will you change therapy?** | • De-escalate to the narrowest active agent: eg Ceftriaxone for susceptible MSSA/H. influenzae/Enterobacterales<br>• Pseudomonas: one active agent based on susceptibility<br>• MRSA: stop Vancomycin/Linezolid if nasal PCR and cultures are negative<br>• Reassess at 48–72 hours<br>• MDR gram-negative VAP (reserve with infectious diseases approval): Ceftolozane-tazobactam, Ceftazidime-avibactam, Imipenem-cilastatin-relebactam or Cefiderocol according to the resistance mechanism |
| **Also be ready for (exam add-ons)** | • Goals: eradicate the pathogen, shorten duration (7 days), avoid resistance<br>• Non-pharm: head of bed 30–45°, subglottic suctioning, oral care (routine chlorhexidine not advised, SHEA 2022), daily sedation interruption, early extubation, hand hygiene<br>• First-line: risk-based empiric therapy: Pip-tazo 4.5 g q6h or Cefepime 2 g q8h or Meropenem 1 g q8h ± Vancomycin/Linezolid for MRSA risk; de-escalate by culture (e.g., to Ceftriaxone)<br>• Pregnancy: β-lactams are safe; avoid aminoglycosides, fluoroquinolones<br>• Renal/dialysis: adjust β-lactams; Vancomycin AUC; Liver: Linezolid caution<br>• Counseling: n/a in ICU; family education on infection control |

## Case 3: Aspiration Pneumonia

**Case details:** Aspiration pneumonia.

| Question | Answer |
|---|---|
| **Q1. Risk factors in this patient?** | • Dysphagia (stroke, neurologic disease)<br>• Decreased consciousness (alcohol, sedatives, seizure)<br>• GERD<br>• NG tube or enteral feeding<br>• Supine position, mechanical ventilation<br>• Poor oral hygiene, dementia |
| **Q2. Treatment regimen and dose (adjust per renal), and what organism will you cover?** | • Typical organisms: oral flora and anaerobes (Peptostreptococcus, Fusobacterium, Bacteroides, viridans streptococci) plus gram-negatives in hospitalized patients; extra anaerobic coverage is only needed for lung abscess or empyema (ATS/IDSA 2019)<br>• First choice: Ampicillin-sulbactam 3 g IV q6h<br>  ◦ Renal: CrCl 15–29 → 3 g q12h; CrCl <15 → 3 g q24h<br>• Alternatives: Ceftriaxone 2 g IV q24h + Metronidazole 500 mg q8h; Moxifloxacin 400 mg IV/PO q24h; Clindamycin 600 mg IV q8h<br>• Duration: 5–7 days; 2–6 weeks for lung abscess or empyema<br>• Non-pharm: swallow evaluation, upright feeding, oral care, treat GERD<br>• Monitor: temperature, oxygenation, WBC, chest X-ray, renal function |
| **Also be ready for (exam add-ons)** | • Goals: treat the infection, prevent further aspiration, avoid unnecessary anaerobic coverage<br>• Non-pharm: aspiration precautions, head of bed elevation, speech therapy/swallow evaluation, oral care, treat dysphagia causes, review sedatives<br>• First-line: Ampicillin-sulbactam 3 g IV q6h (renal adjust); anaerobic coverage only for abscess/empyema (Clindamycin 600 mg q8h or Metronidazole); Penicillin allergy: Levofloxacin or Moxifloxacin<br>• Pregnancy: Ampicillin-sulbactam is safe; avoid fluoroquinolones<br>• Renal: CrCl 15–29: q12h; Liver: no major change<br>• Counseling: upright meals, oral hygiene, avoid sedatives, swallow therapy |

## Case 4: MRSA Osteomyelitis on Vancomycin (GFR 25)

**Case details:** Osteomyelitis caused by MRSA, on Vancomycin, GFR 25 mL/min.

| Question | Answer |
|---|---|
| **Q1. What is the target level for the pathogen?** | • AUC/MIC 400–600 mg·h/L (MIC taken as 1 mg/L) is the recommended target<br>• Older trough target: 10–20 mg/L for MRSA in general |
| **Q2. What is the target level for the type of infection?** | • Serious MRSA infection (bone, endocarditis, bacteremia, meningitis, pneumonia): AUC 400–600<br>• Older trough target: 15–20 mg/L; trough-only monitoring is no longer recommended |
| **Q3. Proper dose and frequency?** | • Loading dose: 20–25 mg/kg if severe infection<br>• Maintenance: reduce because of clearance; for GFR 25 mL/min typically 10–15 mg/kg q24h–q48h<br>• Example (70 kg): load about 1500 mg, then 1000 mg q24h, adjusted by PK<br>• Method: estimate clearance from CrCl, then confirm with Bayesian or two-level AUC<br>• Duration: at least 6–8 weeks for chronic osteomyelitis; longer if high risk of failure<br>• Surgery: debridement is essential<br>• Oral switch criteria: highly bioavailable agent, susceptible organism, adherence, CRP <2 mg/dL, adequate debridement<br>• Newer option to avoid prolonged IV Vancomycin in renal impairment: Dalbavancin 1500 mg IV on day 1 and day 8 (reduce the dose if CrCl <30 and not on dialysis); oral step-down is also supported (OVIVA) |
| **Q4. Monitoring parameters and frequency?** | • SCr, BUN, urine output: 2–3 times/week (daily if unstable)<br>• Levels (AUC): after the 2nd–3rd dose or at steady state; then weekly if stable<br>• CBC: neutropenia and thrombocytopenia with therapy >2 weeks<br>• ESR/CRP: weekly for response<br>• Infusion reaction: infuse ≤1 g/h<br>• Hearing, wound and clinical response<br>• Avoid other nephrotoxins |
| **Q5. When do you need to order another level?** | • Signs of toxicity (rising SCr)<br>• Signs of treatment failure<br>• Change in renal function or fluid status (SCr change ≥0.5 mg/dL or 50%)<br>• Interacting or new nephrotoxic drugs<br>• Dose change, dialysis, hypoalbuminemia |
| **Q6. If the level is >20, what would you do?** | • Hold the next dose(s) and recheck the level<br>• Resume at a lower dose or longer interval once the level is in range<br>• Check SCr and look for AKI; avoid other nephrotoxins<br>• If AUC >600: reduce the dose by 20–25% or extend the interval |

| Question | Answer |
|---|---|
| **Q7. If the level is <10, what would you do?** | • Verify: timing of the sample, adherence, correct dose given<br>• If true low: increase the dose by 20–25% or shorten the interval; consider a loading dose<br>• Check for improved renal function or augmented clearance<br>• Recheck at steady state |
| **Also be ready for (exam add-ons)** | • Goals: eradicate bone infection (6 weeks), achieve target exposure, avoid nephrotoxicity<br>• Non-pharm: surgical debridement, culture-guided therapy, offloading, glucose control, smoking cessation, wound care<br>• First-line: MRSA: Vancomycin AUC 400–600 (trough 15–20 historical), 6 weeks; oral options after response (Linezolid, TMP-SMX + Rifampin, Doxycycline); GFR 25: about 10–15 mg/kg q24h to q48h guided by AUC<br>• Pregnancy: Vancomycin acceptable; avoid fluoroquinolones, tetracyclines, TMP-SMX (first trimester)<br>• Renal/dialysis: level-guided dosing; avoid nephrotoxins; Liver: none<br>• Counseling: complete the course, report rash/hearing loss/low urine output, wound care<br>• Monitoring: Vancomycin level after the 3rd or 4th dose and when SCr changes, weekly SCr/CBC, ESR/CRP, LFTs if Rifampin |

## Case 5: Latent TB with Hepatotoxicity (Anxious Patient) ★ Repeated case, also in 2022–2023, 2021

**Case details:** Latent TB on Isoniazid with malaise, yellow discoloration and AST 154. The patient is frightened and a bit aggressive.

| Question | Answer |
|---|---|
| **Q1. Discuss the hepatic toxicity with the patient, how to manage it, and calm him.** | • Communication: introduce yourself, sit down, acknowledge his fear, listen without interrupting, avoid jargon, use teach-back<br>• Explain: Isoniazid liver injury occurs in 1–2% and usually reverses when the drug is stopped<br>• Assess now: symptoms, AST/ALT, bilirubin, ALP, INR, alcohol, Acetaminophen, herbals, hepatitis serologies<br>• Plan: stop Isoniazid now (symptomatic with AST about 4× ULN); do not rechallenge; inform the physician<br>• After recovery: the physician may choose an alternative latent TB regimen, eg Rifampin 4 months<br>• Counsel: avoid alcohol and Acetaminophen, go to the ED if worsening jaundice, dark urine, vomiting, confusion or bleeding<br>• Monitor: LFTs weekly until normal<br>• Document and report the adverse drug reaction |
| **Also be ready for (exam add-ons)** | • Goals: cure, prevent transmission, prevent resistance and relapse, minimize hepatotoxicity<br>• Non-pharm: airborne isolation until 3 negative smears (or 2 weeks effective therapy and clinical improvement), contact tracing, DOT (directly observed therapy), nutrition, stop alcohol, HIV test<br>• First-line: RIPE: Rifampin 10 mg/kg (600 mg), Isoniazid 5 mg/kg (300 mg), Pyrazinamide 25 mg/kg, Ethambutol 15–20 mg/kg × 2 months then Rifampin + Isoniazid × 4 months (6 months total); add Pyridoxine 25–50 mg daily<br>• Pregnancy: RIPE with Pyridoxine is used (Pyrazinamide is recommended by WHO but omitted by US guidance unless HIV or resistance; without it, give 9 months of Isoniazid + Rifampin + Ethambutol); avoid Streptomycin (fetal ototoxicity); breastfeeding is safe<br>• Renal/dialysis: Isoniazid and Rifampin no adjustment; Pyrazinamide and Ethambutol 3 times weekly (25 mg/kg and 15–25 mg/kg after dialysis) if CrCl <30<br>• Liver: hold if ALT ≥5× ULN (≥3× with symptoms) or bilirubin rise; re-challenge Rifampin first then Isoniazid then Pyrazinamide after ALT <2× ULN; alternatives: Levofloxacin, Ethambutol, Amikacin<br>• Interactions: Rifampin induces CYP3A4: Warfarin, DOACs, Tacrolimus, antiretrovirals (use Rifabutin or adjust Dolutegravir to 50 mg BID), azoles (Voriconazole: use Amphotericin), oral contraceptives, steroids<br>• HIV: start ART 2 weeks after TB therapy if CD4 <50 (and within 8 weeks if ≥50); Dolutegravir 50 mg BID with Rifampin<br>• Counseling: take on an empty stomach, adherence for the full course, orange body fluids (Rifampin), vision changes (Ethambutol: stop and call), jaundice/dark urine/nausea (stop and call), gout (Pyrazinamide), neuropathy (Isoniazid: Pyridoxine), avoid alcohol; hormonal contraceptives fail<br>• Monitoring: LFTs baseline, 2, 4, 8 weeks (monthly if risk), visual acuity and color vision monthly (Ethambutol), uric acid, sputum at 2 months, weight |

## Case 6: CAP with COPD and Penicillin Allergy ★ Repeated case, also in 2022–2023

**Case details:** OSCE with a standardized patient: take the history. CAP with COPD and penicillin allergy; high CURB-65, inpatient.

| Question | Answer |
|---|---|
| **Q1. Diagnosis?** | • CAP requiring hospitalization; high CURB-65 (≥3) suggests severe pneumonia and ICU assessment<br>• Severity criteria (IDSA/ATS): major = septic shock on vasopressors or mechanical ventilation; minor = RR >30, PaO2/FiO2 ≤250, multilobar infiltrates, confusion, BUN ≥20, WBC <4000, platelets <100,000, temp <36, hypotension needing fluids (≥3 minor = severe)<br>• History to take: onset, cough, sputum, fever, pleuritic pain, dyspnea, COPD control, recent hospitalization/antibiotics, allergy details (reaction type and timing), exposures, aspiration, vaccines |

| Question | Answer |
|---|---|
| **Q2. Labs and X-ray?** | • Chest X-ray: infiltrate<br>• Pulse oximetry or ABG<br>• CBC with differential, BMP (BUN, SCr), LFTs<br>• Sputum Gram stain and culture, blood cultures ×2 (severe CAP, or treated for MRSA/Pseudomonas)<br>• Urinary Legionella and pneumococcal antigens (severe CAP)<br>• Viral PCR (influenza, COVID-19) |
| **Q3. What to give (doses)?** | • Clarify the allergy first: non-severe reaction (rash) → Ceftriaxone is acceptable (low cross-reactivity)<br>• Severe CAP, non-severe allergy: Ceftriaxone 2 g IV q24h + Azithromycin 500 mg IV q24h<br>• Severe β-lactam allergy (anaphylaxis): respiratory fluoroquinolone Levofloxacin 750 mg IV q24h or Moxifloxacin 400 mg IV q24h plus Aztreonam 2 g IV q8h<br>• Add Pseudomonas coverage (Piperacillin-tazobactam or Cefepime, or Aztreonam if allergic) and MRSA coverage (Vancomycin or Linezolid) only if prior isolation in 12 months or locally validated risk factors<br>• COPD exacerbation: Prednisone 40 mg daily × 5 days and bronchodilators<br>• Non-pharm: oxygen to SpO2 88–92% in COPD, smoking cessation, pulmonary rehab, vaccines<br>• Monitor: temperature, respiratory rate, SpO2, WBC, QTc (FQ, macrolide), clinical stability criteria |
| **Q4. Why? (Common organisms in CAP)** | • S. pneumoniae<br>• H. influenzae, M. catarrhalis (COPD)<br>• Atypicals: Legionella, Mycoplasma, Chlamydophila<br>• S. aureus, Enterobacterales, Pseudomonas (COPD with risk factors), viruses |
| **Q5. How long?** | • Severe CAP: at least 5 days<br>• Non-severe inpatients and outpatients: 3–5 days<br>• Stop when clinically stable: normal vital signs, able to eat, normal mentation<br>• Re-evaluate if no response after 48–72 hours |
| **Also be ready for (exam add-ons)** | • Goals: cure the infection, prevent complications, minimize resistance and QT/drug interactions, 5-day course<br>• Non-pharm: severity assessment (CURB-65, IDSA/ATS), oxygen, hydration, rest, smoking cessation, vaccines (pneumococcal, influenza, COVID-19, RSV), follow-up chest X-ray only if indicated<br>• First-line: inpatient non-severe: Ceftriaxone 1–2 g daily + Azithromycin 500 mg daily (or Doxycycline 100 mg BID); with penicillin allergy: respiratory fluoroquinolone (Levofloxacin 750 mg daily or Moxifloxacin 400 mg) or Doxycycline; severe: β-lactam + macrolide; add MRSA/Pseudomonas coverage only with risk factors (Vancomycin or Linezolid; Cefepime/Pip-tazo)<br>• Pregnancy: Amoxicillin, Ceftriaxone, Azithromycin safe; avoid Doxycycline and fluoroquinolones; Pneumococcal/influenza vaccines safe<br>• Renal/dialysis: Levofloxacin 750 mg q48h if CrCl 20–49; Gentamicin extended-interval by CrCl with levels; Azithromycin no change<br>• Liver: Azithromycin/Moxifloxacin caution in severe disease<br>• Interactions: Azithromycin/quinolones QT prolongation (stop and use Doxycycline); Doxycycline with ferrous sulfate/antacids (separate by 2–3 h); Levofloxacin tendon risk with steroids; ofloxacin anaphylaxis: avoid all fluoroquinolones<br>• Counseling: complete the course, Doxycycline: sun protection, take upright with water, avoid dairy/iron close to the dose; Azithromycin: palpitations; Levofloxacin: tendon pain, dysglycemia<br>• Gentamicin TDM: extended-interval 5–7 mg/kg q24h (trough <1 mg/L), conventional peak 5–10, trough <2; SCr, hearing/vestibular function |

# 2015

## Case 1: RA, TB and HCAP (Oral)

**Case details:** Oral case combining rheumatoid arthritis, TB and HCAP. Patient has hypothyroidism (high TSH).

| Question | Answer |
|---|---|
| **Q1. Diagnosis; management with agents and doses; problem list and management of each.** | • Pneumonia (HCAP): the term was retired in 2016; today treat by MRSA/Pseudomonas risk factors<br>  ◦ Pip-tazo 4.5 g IV q6h or Cefepime 2 g IV q8h, plus Vancomycin or Linezolid if MRSA risk<br>• TB: Isoniazid + Rifampin + Pyrazinamide + Ethambutol + Pyridoxine<br>• RA on Etanercept: hold the TNF inhibitor during active TB<br>• Hypothyroidism: Levothyroxine; Rifampin increases requirements, so recheck TSH in 4–6 weeks; separate from iron and calcium<br>• Monitor: LFTs, TSH, sputum cultures, visual acuity, RA activity |
| **Q2. RA (Etanercept) and TB management: when to start?** | • Screen before any TNF inhibitor: IGRA/TST and chest X-ray<br>• Latent TB: start latent TB treatment about 1 month before the TNF inhibitor<br>• Active TB: hold the TNF inhibitor; resume after at least 2 months of effective TB therapy and clinical improvement, with rheumatology and ID<br>• Meanwhile: control RA with non-biologic therapy (low-dose steroid, DMARD) as needed |

| Question | Answer |
|---|---|
| **Q3. TB medications with borderline or 3× ULN liver enzymes: what to monitor?** | • Baseline LFTs for all; monitor ALT/AST and bilirubin every 2 weeks for the first 2 months, then monthly, if abnormal baseline<br>• Stop ALT ≥5× ULN (any symptoms) or ≥3× with symptoms<br>• Consider a regimen with fewer hepatotoxic drugs if baseline is already high<br>• Exclude alcohol, Acetaminophen, viral hepatitis<br>• Symptom review at every visit |
| **Q4. Role of vitamin B6?** | • Pyridoxine 25–50 mg daily with Isoniazid<br>• Prevents: peripheral neuropathy and CNS toxicity from Pyridoxine depletion<br>• Higher risk: diabetes, alcohol, malnutrition, renal failure, HIV, pregnancy |
| **Q5. What to monitor with the TB regimen?** | • LFTs and bilirubin<br>• Sputum smear and culture monthly until negative<br>• Visual acuity and color vision (Ethambutol)<br>• Uric acid (Pyrazinamide)<br>• Neuropathy symptoms (Isoniazid)<br>• CBC and platelets (Rifampin)<br>• Drug interactions (Rifampin with Warfarin, Levothyroxine, steroids, contraceptives)<br>• Adherence (directly observed therapy), weight |
| **Q6. If you stop one of them, what is the alternative, and the disadvantages?** | • Stop Isoniazid: Rifampin + Ethambutol + Pyrazinamide ± Levofloxacin or Moxifloxacin for 6–9 months<br>• Stop Rifampin: Isoniazid + Ethambutol + a fluoroquinolone for 12–18 months (longer, less sterilizing)<br>• Stop Pyrazinamide: Isoniazid + Rifampin for 9 months (+ Ethambutol for the first 2 months)<br>• Stop Ethambutol: continue the other three if the isolate is susceptible<br>• General disadvantage: longer duration, higher relapse risk, more toxicity and cost |
| **Also be ready for (exam add-ons)** | • Goals: cure, prevent transmission, prevent resistance and relapse, minimize hepatotoxicity<br>• Non-pharm: airborne isolation until 3 negative smears (or 2 weeks effective therapy and clinical improvement), contact tracing, DOT (directly observed therapy), nutrition, stop alcohol, HIV test<br>• First-line: RIPE: Rifampin 10 mg/kg (600 mg), Isoniazid 5 mg/kg (300 mg), Pyrazinamide 25 mg/kg, Ethambutol 15–20 mg/kg × 2 months then Rifampin + Isoniazid × 4 months (6 months total); add Pyridoxine 25–50 mg daily<br>• Pregnancy: RIPE with Pyridoxine is used (Pyrazinamide is recommended by WHO but omitted by US guidance unless HIV or resistance; without it, give 9 months of Isoniazid + Rifampin + Ethambutol); avoid Streptomycin (fetal ototoxicity); breastfeeding is safe<br>• Renal/dialysis: Isoniazid and Rifampin no adjustment; Pyrazinamide and Ethambutol 3 times weekly (25 mg/kg and 15–25 mg/kg after dialysis) if CrCl <30<br>• Liver: hold if ALT ≥5× ULN (≥3× with symptoms) or bilirubin rise; re-challenge Rifampin first then Isoniazid then Pyrazinamide after ALT <2× ULN; alternatives: Levofloxacin, Ethambutol, Amikacin<br>• Interactions: Rifampin induces CYP3A4: Warfarin, DOACs, Tacrolimus, antiretrovirals (use Rifabutin or adjust Dolutegravir to 50 mg BID), azoles (Voriconazole: use Amphotericin), oral contraceptives, steroids<br>• HIV: start ART 2 weeks after TB therapy if CD4 <50 (and within 8 weeks if ≥50); Dolutegravir 50 mg BID with Rifampin<br>• Counseling: take on an empty stomach, adherence for the full course, orange body fluids (Rifampin), vision changes (Ethambutol: stop and call), jaundice/dark urine/nausea (stop and call), gout (Pyrazinamide), neuropathy (Isoniazid: Pyridoxine), avoid alcohol; hormonal contraceptives fail<br>• Monitoring: LFTs baseline, 2, 4, 8 weeks (monthly if risk), visual acuity and color vision monthly (Ethambutol), uric acid, sputum at 2 months, weight<br>• Biologics: screen TB (IGRA) before any TNF inhibitor; latent TB: treat ≥1 month before starting; active TB: complete the intensive phase first |

## Case 2: Internal Medicine Long Case (TB Component)

**Case details:** Internal medicine case with liver cirrhosis, HCV, DVT prophylaxis and TB.

| Question | Answer |
|---|---|
| **Q1. TB medications: doses and number of tablets per day.** | • Daily doses (56–75 kg): Isoniazid 300 mg, Rifampin 600 mg, Pyrazinamide 1500 mg, Ethambutol 1200 mg, Pyridoxine 25–50 mg<br>• Separate tablets: about 9 per day (Isoniazid 300 mg × 1, Rifampin 300 mg × 2, Pyrazinamide 500 mg × 3, Ethambutol 400 mg × 3) plus Pyridoxine<br>• Fixed-dose 4-in-1 tablet (Rifampin 150, Isoniazid 75, Pyrazinamide 400, Ethambutol 275): 30–37 kg 2 tablets, 38–54 kg 3, 55–70 kg 4, ≥71 kg 5 daily<br>• Cirrhosis/HCV: higher hepatotoxicity risk; consider a regimen with fewer hepatotoxic drugs (Pyrazinamide is the one to avoid in advanced cirrhosis)<br>• Monitor: LFTs every 1–2 weeks at first, then monthly |
| **Also be ready for (exam add-ons)** | • Goals: cure, prevent transmission, prevent resistance and relapse, minimize hepatotoxicity<br>• Non-pharm: airborne isolation until 3 negative smears (or 2 weeks effective therapy and clinical improvement), contact tracing, DOT (directly observed therapy), nutrition, stop alcohol, HIV test<br>• First-line: RIPE: Rifampin 10 mg/kg (600 mg), Isoniazid 5 mg/kg (300 mg), Pyrazinamide 25 mg/kg, Ethambutol 15–20 mg/kg × 2 months then Rifampin + Isoniazid × 4 months (6 months total); add Pyridoxine 25–50 mg daily |

| Question | Answer |
|---|---|
| | • Pregnancy: RIPE with Pyridoxine is used (Pyrazinamide is recommended by WHO but omitted by US guidance unless HIV or resistance; without it, give 9 months of Isoniazid + Rifampin + Ethambutol); avoid Streptomycin (fetal ototoxicity); breastfeeding is safe<br>• Renal/dialysis: Isoniazid and Rifampin no adjustment; Pyrazinamide and Ethambutol 3 times weekly (25 mg/kg and 15–25 mg/kg after dialysis) if CrCl <30<br>• Liver: hold if ALT ≥5× ULN (≥3× with symptoms) or bilirubin rise; re-challenge Rifampin first then Isoniazid then Pyrazinamide after ALT <2× ULN; alternatives: Levofloxacin, Ethambutol, Amikacin<br>• Interactions: Rifampin induces CYP3A4: Warfarin, DOACs, Tacrolimus, antiretrovirals (use Rifabutin or adjust Dolutegravir to 50 mg BID), azoles (Voriconazole: use Amphotericin), oral contraceptives, steroids<br>• HIV: start ART 2 weeks after TB therapy if CD4 <50 (and within 8 weeks if ≥50); Dolutegravir 50 mg BID with Rifampin<br>• Counseling: take on an empty stomach, adherence for the full course, orange body fluids (Rifampin), vision changes (Ethambutol: stop and call), jaundice/dark urine/nausea (stop and call), gout (Pyrazinamide), neuropathy (Isoniazid: Pyridoxine), avoid alcohol; hormonal contraceptives fail<br>• Monitoring: LFTs baseline, 2, 4, 8 weeks (monthly if risk), visual acuity and color vision monthly (Ethambutol), uric acid, sputum at 2 months, weight |

# SOE Bank (year not stated)

## Case 1: Febrile Neutropenia after Stem Cell Transplant (MRSA PCR Positive)

**Case details:** 70-year-old man after hematopoietic stem cell transplantation with persistent fever >38 °C and ANC 98 cells/µL. MRSA PCR is positive.

| Question | Answer |
|---|---|
| **Q1. What is the diagnosis?** | • **Febrile neutropenia** (single temperature ≥38.3 °C or ≥38.0 °C for 1 h; ANC <500 or expected to fall below 500): high risk (post-HSCT, ANC <100, inpatient) |
| **Q2. What is your management plan?** | • Admit; cultures (blood from each lumen and peripheral, urine, sputum if symptomatic), chest imaging<br>• Antipseudomonal β-lactam **within 60 minutes:** Cefepime 2 g IV q8h (or Piperacillin-tazobactam 4.5 g q6h or Meropenem 1 g q8h) by extended infusion, **+ Vancomycin** (MRSA colonization/PCR positive, instability, catheter or skin infection) |
| **Q3. On day 2 he is afebrile and the urine culture is ESBL (sensitive to Piperacillin-tazobactam and carbapenems): what now? Duration?** | • Stop Vancomycin if no gram-positive infection is documented (cultures negative at 48–72 h)<br>• **Meropenem 1 g IV q8h** for ESBL (Piperacillin-tazobactam is inferior for ESBL bloodstream infection: MERINO)<br>• Duration: continue until ANC >500 and afebrile ≥48 h, and complete the infection's own course (complicated UTI 7 days); whichever is longer |
| **Q4. How do you prevent persistent neutropenia?** | • G-CSF: **Filgrastim 5 mcg/kg/day SC** (usual 300 mcg or 480 mcg by weight) after transplant per protocol until ANC recovery (≥1000 for 2–3 days); not routine for established febrile neutropenia, but considered for sepsis, pneumonia, invasive fungal infection or ANC <100 |
| **Q5. Five days later he develops oral thrush with mucositis: treatment?** | • Mild: clotrimazole 10 mg troche 5 times daily or Nystatin 400,000–600,000 units QID (swish and swallow) × 7–14 days<br>• Moderate–severe or esophageal: **Fluconazole 100–200 mg daily** (400 mg for esophageal) × 7–21 days; refractory: Itraconazole solution or Posaconazole; IV Micafungin 100 mg daily if unable to take orally |
| **Q6. What are the risk factors for fungal infection?** | • Prolonged neutropenia (ANC <500 for >7–10 days), broad-spectrum antibiotics, mucositis, central venous catheter, corticosteroids/GVHD, TPN, Candida colonization |
| **Q7. Blood culture grows Candida krusei: plan?** | • C. krusei is **intrinsically resistant to Fluconazole:** switch to an **echinocandin**: Micafungin 100 mg IV daily, or Caspofungin 70 mg load then 50 mg daily, or Anidulafungin 200 mg load then 100 mg daily<br>• Remove or replace the central venous catheter, dilated eye examination, repeat blood cultures daily until negative<br>• Duration: **14 days after the first negative blood culture** and resolution of symptoms/neutropenia |
| **Also be ready for (exam add-ons)** | • Goals: resolve fever and infection, protect against sepsis and invasive fungal disease, recover ANC<br>• Non-pharm: neutropenic precautions (private room, hand hygiene), mouth care, central-line care, avoid rectal temperatures and unnecessary instrumentation, nutrition<br>• First-line: Cefepime 2 g q8h or Piperacillin-tazobactam 4.5 g q6h (extended infusion) ± Vancomycin when MRSA colonized, unstable, catheter or skin infection; Meropenem for ESBL; echinocandin for Candida krusei<br>• Pregnancy (general): Cefepime and Meropenem acceptable; avoid Fluconazole, Voriconazole and aminoglycosides; use Amphotericin B (liposomal) if an antifungal is essential; Filgrastim acceptable<br>• Renal/dialysis: Cefepime reduce (neurotoxicity), Meropenem and Vancomycin dose by CrCl and levels; Micafungin and Caspofungin need no renal adjustment<br>• Liver: Caspofungin maintenance 35 mg in Child-Pugh B; Voriconazole dose reduced; Azole hepatotoxicity<br>• HF/QT: avoid fluoroquinolone/azole QT stacking; monitor K and Mg<br>• Counseling: report fever at once, oral hygiene with soft brush and rinses, avoid raw food and crowds, Filgrastim bone pain (Acetaminophen), no live vaccines |